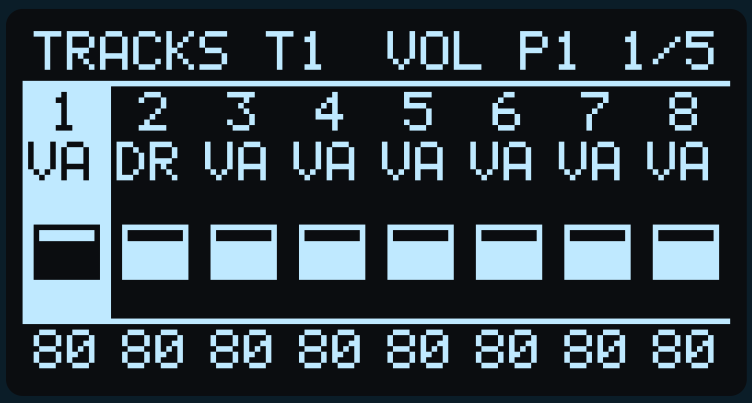
OWNER'S MANUAL

Neptune Groovebox

An eyes-free groovebox played entirely from a Launchkey Mini 25 MK4: eight sound engines, a step sequencer with parameter locks, per-pad pressure, and sampling.



Version 0.9, November 2026

Firmware NPE · macOS

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1 Welcome

Neptune Groovebox turns a Novation Launchkey Mini 25 MK4 into a self-contained instrument. The Mac makes the sound; everything you play, edit, sequence and save is done from the Launchkey, guided by its small display. You never need to look at the Mac.

The design borrows the working habits of hardware grooveboxes: a step sequencer where any step can hold its own value for any knob, a handful of modifier gestures that mean the same thing on every screen, and patterns that you chain into songs.

What is inside

- **Eight sound engines**, one per track: a configurable analog-style synth, six-operator FM, a macro oscillator with eight models, wavetable, wave sequencer, sampler with slicing, time stretch and a granular mode, a 16-voice drum kit, and a MIDI track for outside gear.
- **Five filter models** with a live picture of the response curve.
- **Modulation**: per-voice LFOs with nine shapes and five trigger modes, extra envelopes, and pad pressure routed per track and per note.
- **Effects**: three character blocks on every track (DIRT, TALK, MODFX), insert slots with 30 effect types, two send buses, and a master EQ, compressor and limiter.
- **Sequencer**: 16 patterns, per-track length and speed, parameter locks, trig conditions, micro-timing, live recording, song mode.
- **An arpeggiator on every track**, with variation and mutation, and pressure-driven rolls on the pads.
- **Resampling**: record the master or any track into a new sample, then play it, granulate it or turn it into a wavetable.

How to read this manual

Buttons are shown like `Shift`; names you will see on the display are shown like `CUT`. "Top pad 3" means the third pad from the left in the top row. Knobs are numbered 1 to 8 from the left.

Chapter 2 is a guided first session that ends with a short song. Chapters 3 to 12 are reference. Appendix A lists every short name on the display with its full name, where it lives and what it does.

> **STATUS** This manual describes the software as built. Much of it has so far been checked by automated tests rather than by ear and hand; where the instrument and the manual disagree, the instrument is right and the manual wants correcting.

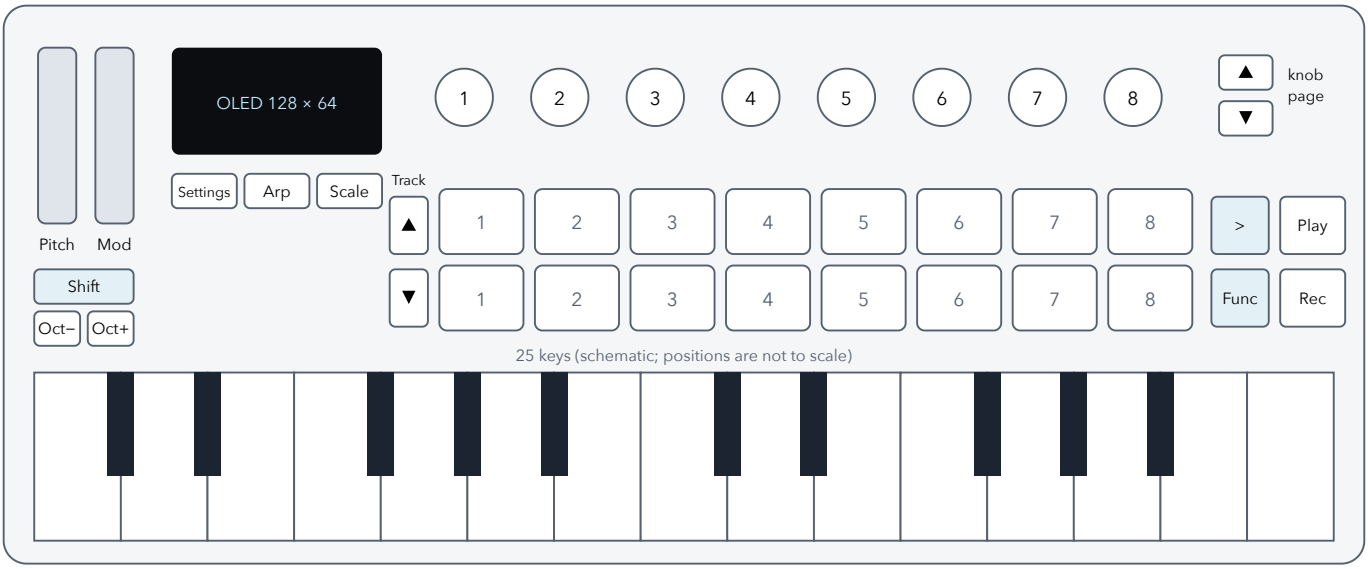
2 Quick start

About half an hour, from silence to a two-pattern song. Each step names the screen you should be on; if you get lost, hold `Shift` and the pads light up as a map of the screens.

> **IF YOU FORGET A COMBINATION, HOLD THE KEY** Hold `Shift`, `Func`, `>`, `▲` or `Rec` by itself for a third of a second and the display turns into a help card: what that key does with the pads, the buttons and the knobs on the screen you are on. The two small knob-page buttons turn the cards. Hold `Func` to read what copy, clear and paste would act on right now, before you press anything. Any other control makes the card step aside and does its job as usual.

Before you start

- Plug the Launchkey into the Mac. It needs the Neptune firmware (tag NPE), which is already on your unit.
- Start *Neptune Groovebox*. The display shows the `TRACKS` screen within a couple of seconds.
- Use headphones or an audio interface if you can. The Mac's built-in speakers add a noticeable delay of their own.
- The app reopens the project you last used. For a clean start: `Settings`, press the lower knob-page button twice to reach `PROJ`, and turn `NEW` (knob 5) four clicks clockwise.



Ten steps to a song

1 Look around

Hold `Shift`. The pads show where each screen lives: the bright pad is where you are. Still holding `Shift`, press top pad 2 to open `TRACKS`, the mixer. Knob 1 is the volume of track 1, knob 2 of track 2, and so on.

The two small buttons beside the knobs turn the knob page on every screen. On `TRACKS` the pages are `VOL`, `PAN`, `SND1`, `SND2` and `ENG`. The pages go round: the upper button on the first page gives the last one, and the lower button on the last page gives the first.

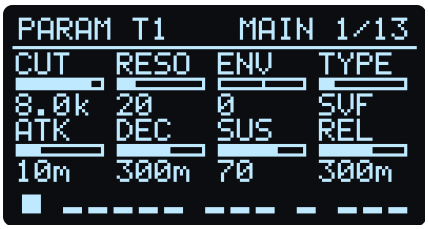
2 Play a sound and choose a preset

Play the keys: track 1 is a synth. Shift + top pad 5 opens PRESET. Scroll with knob 1 (PS) and play: each preset you land on takes over the track's sound, for listening only. Knob 2 (CT) narrows the list to basses, pads, leads and so on. When you like one, turn knob 5 (LD) four clicks clockwise to keep it. Leave without loading and your previous sound comes back.

3 Shape it

Shift + top pad 1 opens PARAM, the pages of the track's sound engine. It opens on MAIN: hold a note and turn CUT (knob 1) and RESO (knob 2), and a picture of the filter's response appears while you turn. Knobs 5 to 8 are the envelope (ATK DEC SUS REL), with a picture of its shape. The knob-page buttons step on to the engine's other pages; the full filter is on FILTR.

Three speeds work on every knob: plain, Shift held for fine, Func held for coarse jumps to useful values.



PARAM opens on MAIN



FLTR: the same filter, drawn large

4 Record a drum beat

Track 2 is a drum kit. Hold ▼ (the lower of the two Track buttons left of the pads) and tap top pad 2 to select track 2, then release. Shift + bottom pad 2 opens DRUMS: the sixteen pads are sixteen drum voices.

In the starting kit the bottom row holds three kicks (pads 1 to 3), a tom, a snare (pad 5), a clap, and closed and open hats (pads 7 and 8). The top row holds six tuned sub-bass hits and two cymbals.

Hold Rec and press Play: the sequencer starts from the top, recording. The name bar says REC FREE and a red light runs across the pads, one pad per step, while the pads go on playing the drums. Tap a beat: a kick on bottom pad 1, the snare on pad 5, hats on pad 7. The pattern loops after 16 steps and what you played comes round again.

Press Play to stop. Recording ends and STEP opens with the drum track selected: your beat is laid out on the pads, ready for editing. Press Play again to hear it.

Free recording keeps your timing exactly. To tighten it afterwards, raise QN on STEP (knob 3): playback is pulled toward the grid without changing what you recorded. To record straight onto the grid instead, press Rec + Play once more while recording: the name bar changes to REC QUANT.

5 Add a bassline with steps

Hold ▼, tap top pad 1, release: track 1 again. Open STEP (Shift + top pad 4). The sixteen pads are now the sixteen steps, top row first. Tap pads to place notes; a quick tap on a lit step removes it.

To choose a step's note, hold its pad and press a key. Do that for each step and you have a bassline.



STEP

6 Make one step different

This is the heart of the instrument. Go to PARAM and the FILTR page. The pads still show the steps. Hold one lit step and turn CUT: only that step changes, and its pad turns orange. That is a *parameter lock*. Any knob on any sound page can be locked to any step.

To remove one lock, hold the step, hold >, and turn the same knob.

7 Add space

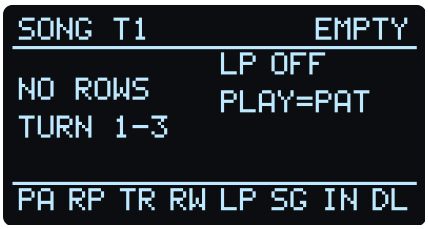
Shift + top pad 3 opens TRK FX. On its first page, SND1 sends the track to a reverb and SND2 to a delay. A little reverb on the snare track, a little delay on the bass.

8 Make a second pattern

Open TRACKS. Press Func + Rec to copy the whole pattern. Hold > and tap top pad 2 to move to pattern 2, then press Func + > to paste. Now change something in pattern 2: remove some bass steps, lock a few filter values, mute the hats by re-recording. Hold > and tap pad 1 or 2 to switch between the two while playing; the change waits for the end of the loop.

9 Chain them into a song

Shift + top pad 6 opens SONG. Turn PA (knob 1): row 1 appears, playing pattern 1. Set RP (knob 2) to 2 repeats. Tap the pad after the last row to add row 2, and set its PA to pattern 2. Turn SG (knob 6) on and LP (knob 5) on, then press Play: the song plays pattern 1 twice, pattern 2, and loops.



SONG

10 Save

Settings, then the PROJ page. Choose a SLOT with knob 1 and turn SVAS (knob 4) four clicks clockwise. The display answers OK. Next time the app starts it reopens this project.

Where to go next

Each of these is a few minutes of play and opens a different side of the instrument.

SQUEEZE Open SCALE (Shift + bottom pad 4), hold a pad and lean into it. Then open AT ROUTE (Shift + top pad 8) and choose what pressure does on this track. Each finger has its own pressure. Chapter 7.

PRESSURE ON ANY KNOB On PARAM, turn a knob, then press Func + ▲. The bottom pad under that knob now sweeps it with pressure and lets go on release. Chapter 7.

EAT YOUR OWN TAIL Open WAVE (Shift + bottom pad 5), go to its REC page and record the master output for a bar. The take becomes a sample you can play on a sampler track. Chapter 11.

FREEZE TIME On a sampler track set MODE to GRAN and SCAN to 0 on the GRAIN page. The sound stands still; move STRT to walk through it by hand. Chapter 11.

ANY LOOP, YOUR TEMPO Put a loop on a sampler track, set MODE to LOOP, then on STRCH choose an ALGO, switch SYNC on and set LEN. Change the tempo: it follows. Try each algorithm on the same loop. Chapter 11.

CHOP A BREAK On SLICE set MODE to TRNS and PLAY to KEYS: every hit of the loop is now on its own key. Or PLAY = RAND with steps on every sixteenth. Chapter 11.

A WAVETABLE FROM ANYTHING On a wavetable track set TABL to USER, then choose a recording with SM on the WAVE screen's REC page. POS now scans through your recording, and the SCAN page can play it back at any length. Chapter 5.

THREE WAVES, ONE TABLE On a wavetable track open WAVE: give key 1 a sample with SM, press the second top pad and give key 2 another, then a third. Set AL to TRNS or VOC and sweep POS. Chapter 5.

PATTERNS THAT NEVER REPEAT Give two tracks different lengths (LN on STEP): 16 against 12, say. Then give some steps a condition (COND: PROB or A:B). Chapter 9.

AN ARP THAT EVOLVES Press Arp twice and hold a chord. On the VARY page raise SPCE; on MUTA switch mutation on and let it run. When you like what it has become, >PAT writes it into the pattern. Chapter 10.

ROLLS UNDER YOUR FINGER On the ARP screen's ROLL page switch ROLL on. Now a held drum pad repeats, faster the harder you press. Chapter 10.

A SYNTH THAT TALKS TRK FX, TALK page: raise MIX, sweep VOWL, add DPTH. Chapter 8.

PUMPING Put the DUCK effect on a bass or pad track and set its SRC to the drum track. Chapter 8.

EVERY NOTE SLIGHTLY DIFFERENT On the synth's ANALG page raise SLOP, FVAR and EVAR. On drums, point an LFO at TONE with TRIG = HOLD and SHPE = RND. Chapters 5 and 7.

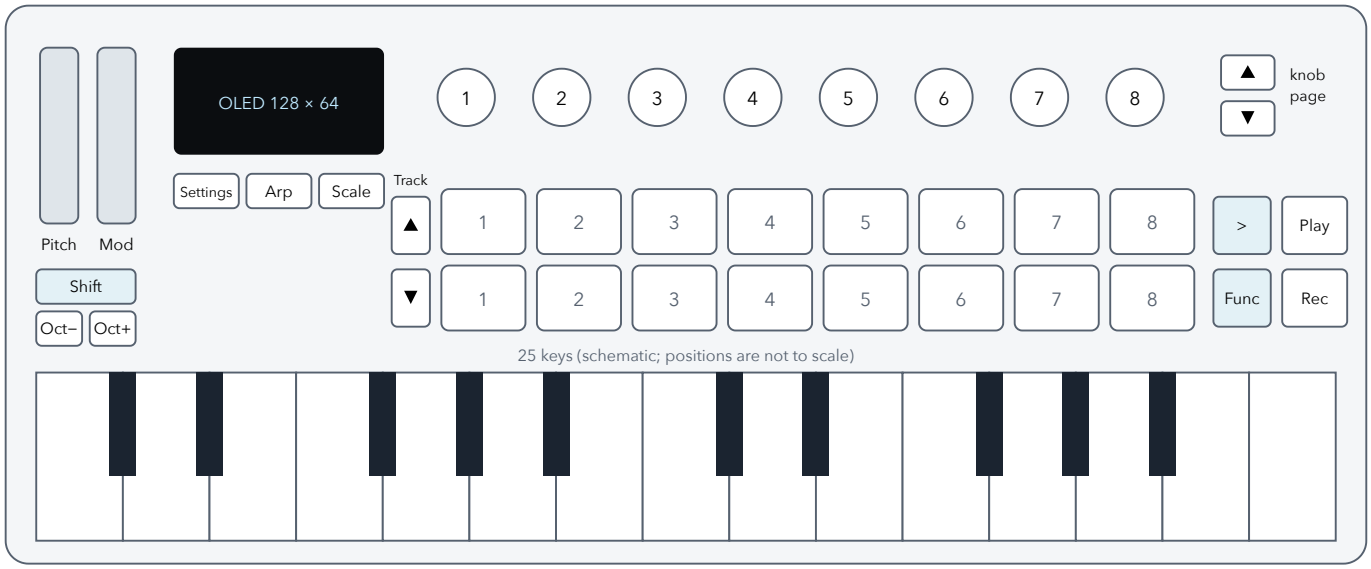
JAM WITHOUT FEAR Hold > and press Rec to keep a copy of the pattern. Wreck it. Hold > and press Play to bring it back. Chapter 9.

ALL TRACKS AT ONCE Hold ▲ or ▼ and turn a knob: the same control moves on every track that has it. Sweep every filter together. Chapter 3.

TWO HANDS, TWO REGISTERS `Shift` + `Oct−` drops the lowest octave of the keyboard by a further octave or more: bass in the left hand, chords in the right. Chapter 12.

3 Finding your way

The control surface



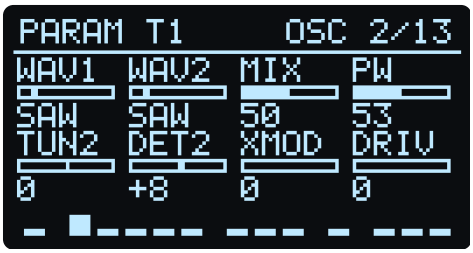
- **Eight endless knobs.** They always edit the eight things shown on the display, in the same positions.
- **Knob-page buttons** (the small pair right of the knobs): previous and next page of knobs, on every screen. The pages go round, so from the first page the upper button reaches the last in one press.
- **Sixteen pads**, velocity- and pressure-sensitive. What they do depends on the screen: play notes, hold steps, select rows.
- ▲ ▼ left of the pads (printed *Track*): tapped, they change what the pads address (the next sixteen steps, the next scale page, the next row). Held, they open the Track layer.
- `>` right of the top pad row: the pattern key.
- `Func` right of the bottom pad row: "bigger, or secondary".
- `Shift`: "where, and how finely".
- **Any of these held alone** (`Shift`, `Func`, `>`, ▲ or ▼, `Rec`) shows its help cards on the display (see *Help cards* below).
- `Settings` `Arp` `Scale` `Play` `Rec` do what they say. `Oct−` `Oct+` shift the keys.
- **Pitch strip and mod strip** act on the selected track.

The display

The top line is the name bar: the screen or page name on the left, the selected track (`T 1`) on the right, and tags when something is engaged: `ARP` , `LATCH` , `PIN` , `SPLIT` .

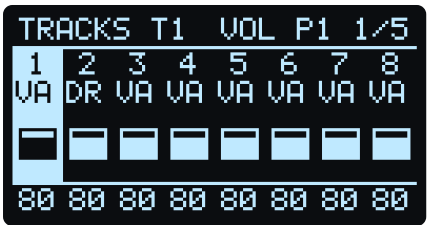
Grid screens show eight cells in two rows of four. Knobs 1 to 4 are the top row, 5 to 8 the bottom row.

Bar screens show a picture or list with a strip of eight two-letter labels along the bottom, one per knob. When you turn a knob the name bar

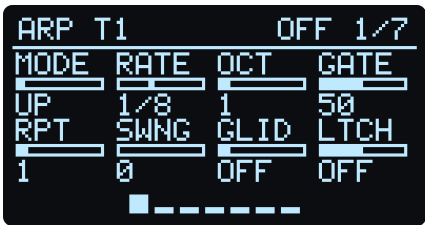


A grid screen: the synth's OSC page

spells out its full name and value. A knob that has nothing to act on at the moment says why there for a second instead of doing nothing: `SAMPLER ONLY`, `NOT ZOOMED` and `ONE SLICE` on `WAVE`, `ROUTE OFF` on `AT ROUTE`, `FLT B IS OFF`, `FILTER IS OFF` and `SEM MRPH ONLY` on the filter pages, `SLOT IS EMPTY` on `TRK FX`, `NOTHING KEPT` and `NOTHING TAPPED` on `ARP`, `SONG IS EMPTY` and `ONE ROW` on `SONG`, `NO LOCKS` on a held step.



A bar screen: TRACKS, VOL page



ARP, first page

Screens: Shift + pad

Hold `Shift` and press a pad. The screens follow the legends printed above and below the pads.

PAD	TOP ROW	BOTTOM ROW
1	`PARAM` – the sound engine's pages	`STEP` – the step sequencer
2	`TRACKS` – mixer and engine choice	`DRUMS` – drum pads
3	`TRK FX` – track effects	`ARP` – arpeggiator and rolls
4	`STEP` – the step sequencer	`SCALE` – scale pads
5	`PRESET` – sound presets	`WAVE` – waveform and resampling
6	`SONG` – song rows	`FLTR` – the filter picture
7	`SETTINGS` (also the `Settings` button)	–
8	`AT ROUTE` – pad pressure routing	–

The grammar: one gesture, one meaning

These hold on every screen. Learn this table and you know the instrument.

HOLD	AND TURN A KNOB	AND PRESS A PAD	AND PRESS ▲ / ▼	AND PRESS A KNOB-PAGE BUTTON
nothing	edit, one step	the screen's pad action	what the pads address: step page, scale page, song row, preset list	knob page ±1
`Shift`	fine: a tenth of a step (on an envelope's `ATK`, `DEC` or `REL`: that stage's curve, see Filters)	go to that screen	previous / next track	–
`Func`	coarse: jump to useful values	top row: jump to a page; bottom row on `TRACKS`: solo	set the low (▼) or high (▲) pressure-range end of the knob turned last	section jump, or first / last page

HOLD	AND TURN A KNOB	AND PRESS A PAD	AND PRESS ▲ / ▼	AND PRESS A KNOB-PAGE BUTTON
a step pad	lock that knob on that step	–	–	lock page ±1
`>`	on a held step: remove that knob's lock	queue pattern 1–16	–	(`Rec` keeps a copy of the pattern, `Play` reloads it)
`▲` or `▼`	the same control on every track	top row: select track N; bottom row: mute track N	–	–

Knob speeds

Plain is 1 % of a continuous range or one item of a list. `Shift` is a tenth of that. `Func` snaps: levels to 0, 25, 50, 75, 100 % and the default; two-sided values to the ends, the centre and the quarters; tuning and frequencies by octaves; tempo by 10; lengths by 4; division lists to the straight (not dotted or triplet) values; an LFO's `DEST` list from one page's group of destinations to the next.

Buttons

BUTTON	PLAIN	WITH SHIFT	WITH FUNC
`Arp`	Opens `ARP`; pressed again there, switches the track's arp on or off	Latch	–
`Scale`	Scale lock on / off (the name bar says which, with the scale)	Opens `SCALE`	–
`Settings`	Opens `SETTINGS`	–	Tap tempo
`Play`	Start / stop; stopping also ends recording	Stop and return to the start	Clear
`Rec`	Recording on / off, when the button is let go. Held + `Play`: start recording; again: free / quantized	Fill, while held	Copy
`>`	Pattern key (hold)	–	Paste

Help cards

Every key that changes what the other controls do while you hold it has help cards: `Shift`, `Func`, `>`, `▲` / `▼` and `Rec`. Hold one of them by itself. After a third of a second the display becomes a help card for that key, and stays for as long as you hold it. While a card is showing, the two knob-page buttons turn to the key's other cards (the mark at the top right, such as `2/3`, says which one you are on). Anything else you press or turn makes the card step aside for the rest of that hold and then does exactly what it always does, so a quick `Shift` + pad or `Func` + `Play` never sees a card at all. Let go and the screen is back as it was.

`Shift` has three cards, the same everywhere: the pads (a map of the screens, the one you are on inverted), the buttons, and the knobs. `Func` has three that follow the screen: *edit* (what `Rec`, `Play` and `>` would copy, clear and paste right now, and what undo and redo would take back or do again), the pads (the page keys by name), and the knobs and other buttons. `HELP` on `SETTINGS` → `SETUP` chooses when the cards appear: `OFF`, `0.35S` or `INST` (at once).

`>` has two cards: the sixteen patterns laid out like the pads (the current one inverted, the one queued next marked `*`, patterns with steps in them by number, empty ones as a dot), and what `Rec`, `Play` and a knob do under it,

including whether Play would reload a kept copy or the pattern as the project had it. ▲ / ▼ have two: the eight tracks over the top pads with their engines and, under each, ON, MUTE or SOLO; and what a knob does under them here and what a quick press of the buttons moves on this screen. Rec has one: what letting go will do, and the chord with Play.

Two things to know. A quick press of ▲ or ▼ still pages as always; a press held long enough to show the card does not page when you let go, because the card took its place. Rec is the exception: it switches recording on or off when you let go whether or not its card was up. Play, Settings, Arp and Scale act the moment you press them and have no cards; what they do with Shift or Func is on those keys' cards.

Copy, clear, paste

Func + Rec copies, Func + Play clears, Func + > pastes. What they act on follows where you are: a held step if you are holding one (STEP : the whole step; PARAM, TRK FX, FLTR : its locks); otherwise the track's steps on STEP, the whole pattern on TRACKS, the track's sound on PARAM and FLTR, and the track's effects on TRK FX. On PARAM and FLTR copy and paste take the whole sound (engine, mixer, effects), and clear resets the engine's own parameters and leaves the mixer and the effects alone, so the display calls it RESET. On TRK FX all three act on the effects only: the DIRT, TALK and MODFX blocks, the insert slots and the ARPLK values. Clearing switches the blocks off, empties the slots and sets the arp locks to off; volume, pan, sends, length and speed on the TRACK page are mixing and sequencing and are never touched. On other screens the three do nothing, and the name bar says so (NO CLEAR HERE). The name bar confirms what was done (CLEAR T3 STEPS, CLEAR T3 FX). If you are not sure what a combination will act on, hold Func and read the card first.

A whole pattern needs a held button. On TRACKS, clear and paste replace every track's steps, so Func + Play and Func + > act only after the button has been held for half a second. The name bar (or the card, if it is showing) fills up while you hold: CLEAR >>>.. . Let go early and nothing happens; the name bar says HOLD TO CLEAR. Everything smaller acts at the press.

Undo and redo

Hold Func and Shift and press Play to undo; hold them and press Rec to redo. The last fifteen actions can be taken back, one press each, and it does not matter what you played, turned or looked at in between: an action stays undoable until it is undone or a sixteenth pushes it out. The name bar says what was taken back (UNDO CLEAR T3 STEPS), and the Func card shows what the next undo and redo would be before you press (UNDO: CLEAR T3 STEPS, REDO: ..., or NOTHING TO UNDO).

These are undo steps: every clear, paste and reset; reloading a pattern with > + Play; changing an insert's TYPE; LCKS clearing a step's locks; EVOL turned back to the ancestor; >PAT ; and a recording, one step from the moment it starts to the moment it stops, so a bad take is one press away from gone (also while it is still running: the take so far goes and recording carries on). Knob turns are **not** undo steps. Undoing a change to a sound puts back what that action changed and leaves knobs you turned since as they are. Loading a project or starting a new one empties the list, and a step whose track has been given another engine since is skipped.

The older shortcut still works: the same copy, clear or paste combination pressed again straight away, with nothing in between, takes it back.

Things that destroy work

Loading, saving over, deleting and the like sit on a knob and need four clicks clockwise, shown as >>> filling up. Any other knob, or a click back, cancels. Nothing destructive is ever on a pad. Clearing or pasting over a whole pattern needs its button held for half a second, and whatever the edit combinations do can be undone.

Page keys

On screens with several pages, `Func` + a top-row pad jumps straight to a page (on `PARAM`, to a section; the same pad again steps within it).

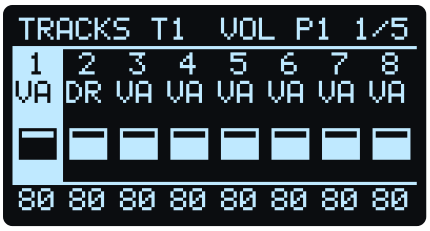
Pinning the playing pads

On `DRUMS` or `SCALE`, hold `Shift` and hold that screen's own bottom pad for half a second. `PIN` appears in the name bar and those pads keep playing, with pressure, on every other screen, so you can play pads with one hand and turn sound knobs with the other. The same gesture unpins.

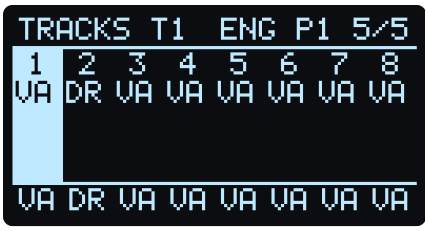
4 The screens

Every screen, what its knobs show and what its pads do. Short names are explained in Appendix A.

TRACKS – the mixer



VOL



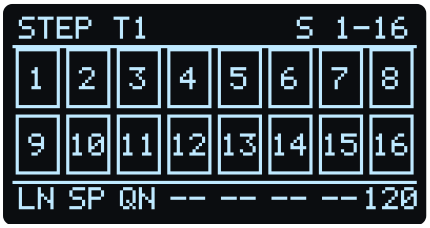
ENG

Knobs: knob N belongs to track N. Five pages: `VOL`, `PAN`, `SND1`, `SND2`, `ENG`.

Pads: the top row selects a track. The bottom row mutes (green = sounding, red = muted, blue = solo); `Func` + bottom pad solos.

Changing a track's engine: on the `ENG` page turn the track's knob to choose an engine. It is shown boxed as pending and nothing changes yet. Tap that track's top pad (lit amber) to commit. Leaving the page drops the choice. Changing engine replaces the track's sound in every pattern.

STEP – the step sequencer



STEP

Knobs: `LN` track length, `SP` track speed, `QN` playback quantize, and the tempo on knob 8.

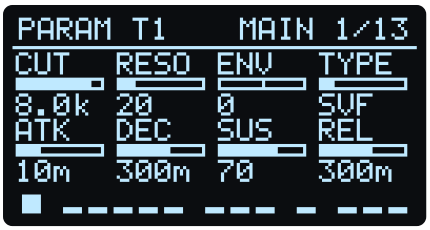
Pads: sixteen steps. A tap adds a step; a quick tap on a lit step removes it; a held step is never removed. `▲` / `▼` move to the next sixteen steps of a longer track. Hold a step and press a key to set its note (several keys for a chord).

Step detail: hold a step (a little longer than a tap) and the screen changes to that step's own pages, without anything else being touched. It opens on the track's `MAIN` page (a drum track: the `DR` page of the drum that step plays; a MIDI track: its first page), and every cell shows the value that step will play: a small mark in the cell's corner means the step has its own value there (a lock), otherwise it is the track's value. Every turn is a lock. The knob-page buttons move through all the track's sound pages while the step is held, and one press back from `MAIN` is `TRIG` (`NOTE VEL LEN MICR` | `COND PROB A B LCKS`), the step itself. The name bar shows the step (`S5 LOCK`). Let go and `STEP` is back. Turning a knob straight after pressing opens the pages at once; a quick tap never shows them. With several steps held, the pages show the one nearest the top left and every held step is edited together: a knob gives all of them the value you see, and a key gives all of them its note.

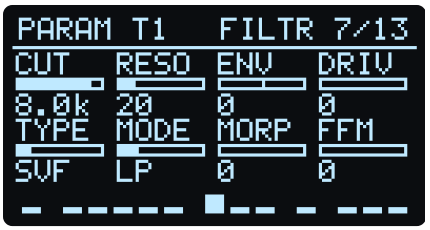
Recorded pressure: a step whose notes were recorded from pads with pressure has a small dark dot at the lower right of its cell, and one more page, `PRES`, one press back from `TRIG` (the pages go round, so it is also the last page). It shows the recorded curves over the note's length and three cells: `PRES` sets how much of the pressure is

played (0 to 200 %; 0 plays none and keeps the recording), `>LCK` turns it into parameter locks, `CLR` removes it. `>LCK` and `CLR` need four clicks clockwise.

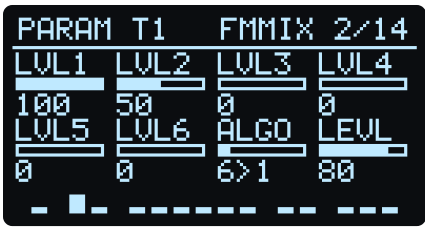
PARAM – the sound



Synth, MAIN (the landing page)



Synth, FILTR

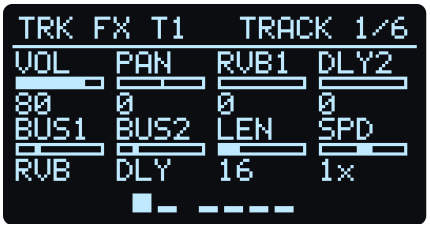


FM, FMMIX

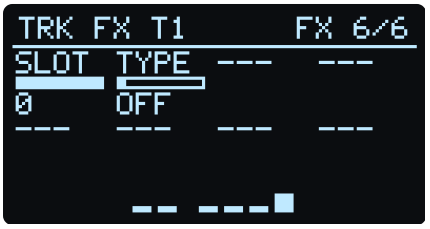
Knobs: the pages of the selected track's engine (Chapter 5), opening on `MAIN`: cutoff, resonance and the level envelope, the same on every engine that has a filter. A drum track shows one `DR` page for the pad last selected on `DRUMS` (and its `KICK` page when that pad is a kick machine), then its LFOs.

Pads: the track's steps, for locks: dim = empty, green = has a note, orange = has locks, white = held. A bottom-row pad under a knob that has a pressure range is violet and plays that range instead.

TRK FX – track effects



TRACK



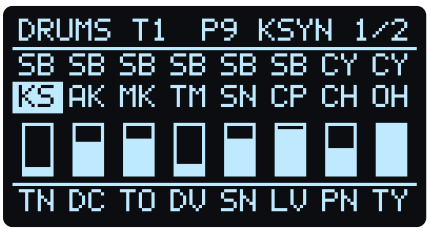
FX

The same six pages on every track, each with a page key (Func + top pad 1 to 6):

- `TRACK`: `VOL PAN SND1 SND2` | `BUS1 BUS2 LEN SPD`. The send cells are labelled by what the bus carries (`RVB1`, `DLY2`).
- `ARPLK`: `GATE VEL RTCH`, there to be locked per step so the arp changes along the pattern.
- `DIRT`, `TALK`, `MODFX`: the three character blocks (Chapter 8).
- `FX`: `SLOT TYPE ON WET` and the chosen insert's four controls.

Pads: as on `PARAM`; every cell here can be locked and given a pressure range.

DRUMS – drum pads

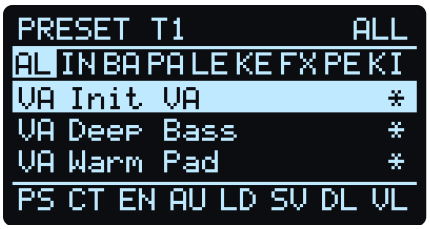


DRUMS

Pads: sixteen drum voices, played with velocity and pressure. One colour per kind of voice; white while it sounds. Playing a pad selects it.

Knobs: the selected pad's voice: `TN DC TO DV SN LV PN TY` (tune, decay, tone, drive, snap, level, pan, type). The screen follows the selected track if it is a drum track, otherwise the first drum track.

PRESET – sound presets



PRESET

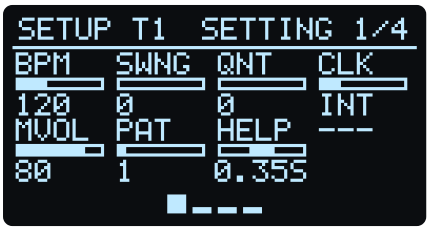
Knobs: `PS` scroll, `CT` category, `EN` this engine or any, `AU` audition, `LD` load, `SV` save, `DL` delete, `VL` track volume. Load, save and delete need the four clicks.

Audition: opening the screen changes nothing. As soon as the highlight moves (knob 1, ▲ / ▼, or a change of category or engine filter), the highlighted preset takes over the track's sound for listening only, and the name bar shows AUD. The track's own sound comes back when you leave the screen by any route or select another track. `LD` keeps the highlighted preset instead (the name bar shows LOAD OK): after that nothing is put back. `AU` is the off switch: turned off (AU- in the name bar), the list scrolls without changing the sound, and `LD` still loads. A project saved while you are listening stores the track's own sound.

Pads: play the track, sixteen semitones up from middle C, so you can audition without reaching for the keys. ▲ / ▼ also scroll the list.

Your own presets are files in *Music / Neptune / Presets*.

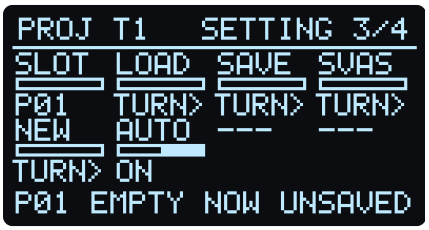
SETTINGS



SETUP



MASTR



PROJ

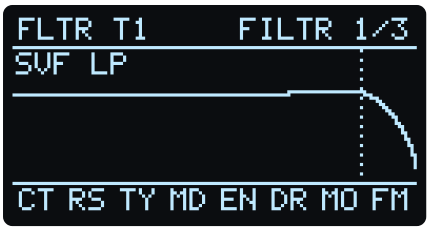


CLICK

Four pages. `SETUP` : tempo, swing, quantize, clock source, master volume, pattern, and `HELP` (when the help cards appear). `MASTR` : master EQ, compressor and limiter. `PROJ` : project slots (Chapter 12). `CLICK` : the metronome – `CLCK` (when it clicks), `CLVL` (how loud) and `CNT` (the count-in before a recording); see Recording in the sequencer

chapter. The metronome's settings belong to the instrument, not to a project: loading or starting a project leaves them as they are, and they are kept between sessions.

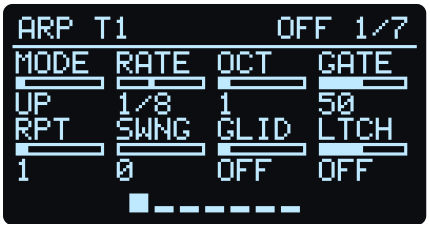
FLTR – the filter picture



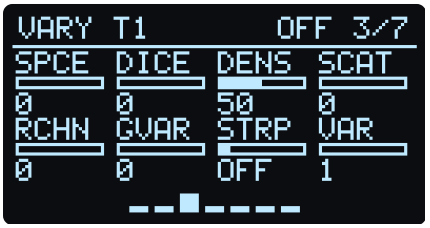
FLTR

The filter's actual response, measured from the filter itself, with the engine's filter controls beneath as two-letter labels. For the synth's two filters it shows them combined as routed. Further filter pages of the engine follow on the knob-page buttons. On the filter envelope's page, turning an envelope knob shows the envelope's shape in place of the response for a second. Drums and MIDI tracks have no filter.

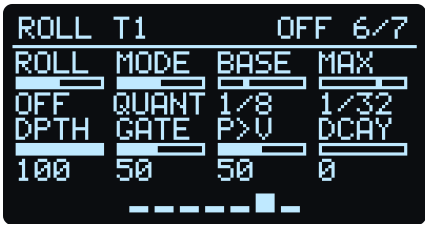
ARP – arpeggiator and rolls



ARP



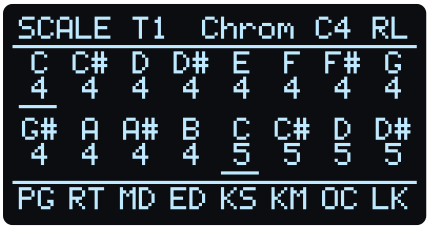
VARY



ROLL

Seven pages for the selected track's arpeggiator: `ARP`, `PLAY`, `VARY`, `MUTA`, `ACFG`, and `ROLL` / `ROL2` for pad rolls (Chapter 10). An `ARP` tag in the name bar shows the arp is on.

SCALE – scale pads

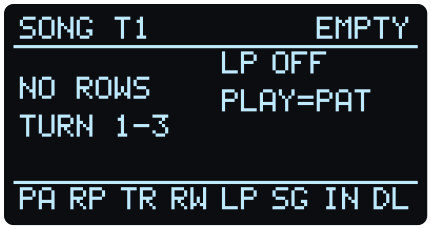


SCALE

Pads: play the selected track in a scale, with pressure; the root is blue, other notes green. ▲ / ▼ change the scale page. Twelve pages are built in (chromatic, major, minor, pentatonics, blues, whole-tone and modes) and you can make your own.

Knobs: `PG` page, `RT` root, `MD` relative or absolute, `ED` edit, `KS` and `KM` let the keys choose pages, `OC` octave, `LK` scale lock (the same switch as the `Scale` button: with the lock on, the keys play the nearest note of the page's scale on synth tracks, arps stay in it, and `SCALE` shows in the name bar). In edit mode tap a pad and press a key to assign its note.

SONG

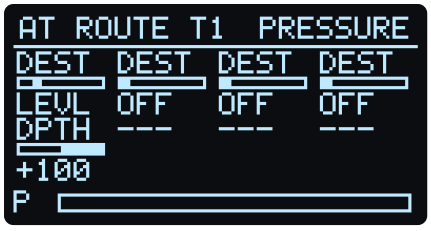


SONG

Knobs: `PA` pattern, `RP` repeats, `TR` transpose, `RW` row, `LP` loop, `SG` song on / off, `IN` insert a row, `DL` delete the row (four clicks).

Pads: pad N selects row N; the pad after the last row adds one. With `SG` on, `Play` follows the rows; a row ends when track 1 has wrapped the set number of times.

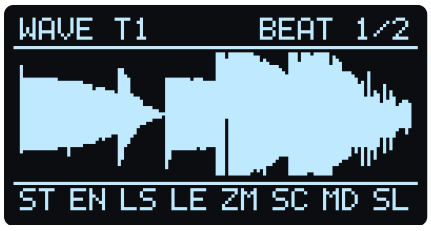
AT ROUTE — pad pressure



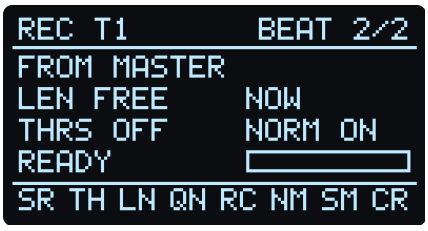
AT ROUTE

What pad pressure does on the selected track. Top row: the destination of routes 1 to 4. Bottom row: how far each moves. Every track has its own routes; the destinations on offer depend on the engine (Chapter 7). The pads play the track chromatically so you can feel the result.

WAVE — waveform and resampling



WAVE



REC

`WAVE` page: the track's sample with start and end brackets, loop marks, slice ticks and a moving line for each sounding voice (in granular mode, a tick per grain). Knobs: `ST EN LS LE` markers, `ZM` zoom, `SC` scroll, `MD` play mode, `SL` slices. Pad N selects slice N.

On a wavetable track the page is the key-frame editor of the `KEYS` table (Chapter 5): the sample of one key, a line where its cycle is taken and a tick for every other key on the same sample. Knobs: `KY` key, `SM` its sample, `PO` its position in the sample, `PL` its place along the table, `ZM` zoom, `SC` scroll, `AL` algorithm, `TB` table. The top-row pads choose the key; the lit ones are in use.

`REC` page: resampling (Chapter 11).

5 Sound engines

Each track has one engine, chosen on `TRACKS` → `ENG`. Its pages appear on `PARAM`. All engines except drums and MIDI have a filter (Chapter 6); all sound engines have LFOs (Chapter 7).

The landing page: MAIN

`PARAM` opens on the same page for every engine that has a filter, so the first handles of a sound are always under the same knobs: knob 1 is the filter cutoff (`CUT`), knob 2 the resonance (`RESO`), knob 3 how far the filter envelope moves the cutoff (`ENV` ; on FM, macro, wavetable, wave sequencer and synth; the sampler shows `ENVF`), knob 4 the filter `TYPE` , and knobs 5 to 8 are the envelope that shapes the level: `ATK` , `DEC` , `SUS` , `REL` (on FM, operator 1's). These are the same cells as on the engine's `FILTR` and `AMP` pages further on: a value, a parameter lock or a pressure range set here is set there too. Turning them shows the filter or envelope picture. The engine's own pages follow, in the tables below.

How the engines relate

The engines are separate instruments, one per track. The synth is the one that mixes kinds of oscillator: each of its three oscillators can be an analog wave, a wavetable, an FM pair, a formant oscillator or a harmonic oscillator, in any combination, into the same mixer, filters and envelopes (see the synth, below). The FM, macro, wavetable and wave-sequencer engines go deeper into one idea each (six operators, physical models, timed scans through a recording, sequenced waves) and are chosen per track. `ARCH` (mono, hybrid, poly) only loads different starting values into the synth's own pages. Every engine shares what comes after the oscillators: the same five filter models, the same envelopes and LFOs, the same effects and the same sequencer.

There is also a bridge between engines: resampling. Any track (a macro oscillator, an FM sound, the synth) can be recorded on the `WAVE` screen's `REC` page, and the recording becomes a sample for a sampler track or the `USER` table of a wavetable track (Chapter 11). That is how a macro-oscillator sound becomes a wavetable. To layer two engines on one note, put them on two tracks and play or sequence both.

Synth – the configurable voice

Three oscillators and noise into two filters and an amplifier, up to sixteen voices. `ARCH` on the `VOICE` page loads a starting point (mono, hybrid, poly) into the same pages; `MODE` sets mono, duo, poly or unison; `STCK` stacks detuned copies. `PW` shapes every waveform. At 50 each wave is plain, and the two directions are two different sounds that grow smoothly out of it:

WAVE	BELOW 50: THINNER, SHARPER	ABOVE 50: THICKER
`SQR` `PLS`	The pulse narrows: classic pulse width, thin and nasal at the bottom.	A square an octave higher is mixed in, up to half and half.
`SIN` `TRI`	Wavefolding: the wave is driven up to six times too hard and the excess folded back, for bright, glassy harmonics.	Saturation: the wave is pushed into a limiter and squares off, warm then hollow.
`SAW`	A second saw is added, a part of a cycle behind the first: the doubled, phasing saw of saw PWM.	A saw half a cycle behind is taken away: the saw hollows out into a square.

An LFO sent to `PW` animates whichever wave is selected; centred on 50 it sweeps through both sounds.

Oscillator kinds: wavetable, FM, formant, harmonic

`WAV1` , `WAV2` and `WAV3` go on past the analog waves to four other kinds of oscillator: `WT` , `FM` , `FORM` and `HARM` . Any slot can be any kind, so a wavetable can sit beside a saw and an FM pair, all three through the same filter. Tuning,

detune, cross-modulation, the stack and the mixer treat every kind alike. Each non-analog slot has two controls of its own on the SRC page, labelled with the slot's number (slot 1 on knobs 1–2, slot 2 on 3–4, slot 3 on 5–6; a slot playing an analog wave shows nothing there):

KIND	FIRST CONTROL	SECOND CONTROL	WHAT IT IS
WT	POS position in the table	TBL table: SAW SQR TRI SIN PWM FORM HARM SPEC DIGI, then VOWL SYNC FOLD FM1 FM3 BELL ORGN NOIS RESO WLSH PRIM OCTV PDST COMB	The wavetable engine's oscillator in a slot: any of its 23 built-in tables, read the same way; sweep POS for movement.
FM	IDX modulation depth	RAT modulator ratio, 1–12	A sine modulated by a second sine at a whole multiple of its pitch: bells, basses, glassy tones.
FORM	FRM formant frequency, 1–16 times the pitch	WID width of the formant	A resonant, vocal peak that stays put while the note changes character around it.
HARM	TLT tilt, bright to dark	ODD all harmonics to odd only	Up to twelve harmonics added together; odd-only is hollow like a square.

Whatever moves PW (an LFO with DEST PW, pad pressure routed to PW) also moves the first control of every non-analog slot, so one LFO can sweep a pulse's width and a wavetable's position together. Choosing a new kind for a slot starts it from that kind's own settings. The wavetable kind is the wavetable engine's own oscillator on the wavetable engine's built-in tables (described in the Wavetable section below), read smoothly. What stays in the wavetable engine: tables made from your own recordings (USER , KEYS), the timed SCAN , and the coarse read-outs of INTP . An LFO sent to a slot's table cell moves it by eight tables at full depth.

PAGE	KNOBS 1–4	KNOBS 5–8
OSC	WAV1 WAV2 MIX PW	TUN2 DET2 XMOD DRIV
OSC3	WAV3 TUN3 LVL3 NOIS	NCOL TUNE FINE BEND
SRC		
VOICE	ARCH MODE VOX LEGA	STCK SPRD GLID GLMD
ANALG	DRFT DRAT SLOP PHAS	WSHP SOFT FVAR EVAR
FILTR	CUT RESO ENV DRIV	TYPE MODE MORP FFM
FLT B	TYPB MODB OFSB RESB	ROUT BAL DRVB TRKB
F ENV	ATK DEC SUS REL	CURV KEY VEL PMOD
AMP	ATK DEC SUS REL	CURV VEL LEVL

Three more pages follow for modulation: MOD and MOD2 (up to four LFOs) and MENV (up to four extra envelopes). See Chapter 7.

The ANALG page is the instrument's "age" control: drift, tuning slop, and per-note variation of waveform, filter and envelope. With everything at zero, successive notes are identical; raise them and no two are.

FM — six operators

Six sine operators, eight algorithms. FMMIX puts all six operator levels under knobs 1 to 6 so you can play the timbre directly; RATIO does the same for their frequency ratios. Each operator then has its own page with ratio, level,

detune, feedback and envelope.

PAGE	KNOBS 1–4	KNOBS 5–8
FMMIX	LVL1 LVL2 LVL3 LVL4	LVL5 LVL6 ALGO LEVL
RATIO	RAT1 RAT2 RAT3 RAT4	RAT5 RAT6 TUNE BEND
OP1–OP6	RATI LEVL DETN FDBK	ATK DEC SUS REL
FILTR	CUT RESO ENV DRIV	TYPE MODE MORP KTRK
F ENV	ATK DEC SUS REL	CURV EVEL VEL WHL
STRIP	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
LFO1	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
LFO2	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Algorithms: 6>1 · 2+3 · 3x2 · 2x3 · 1>3 · ADD · 2+4 · 4+2.

Macro — eight models, four knobs

One oscillator that can be eight different things. MODL chooses the model, and knobs 2 to 5 are renamed to suit it.

MODEL	WHAT IT IS	KNOBS 2–5
VA	virtual analog	SAWS SPRD HARM MORP
FM	two-operator FM	INDX RATI DETN FDBK
FOLD	wavefolder	FOLD SHPE BIAS CHEB
FORM	formant	VOWL SHFT Q MORP
HARM	harmonic	TILT EVOD NUM STRC
PLCK	plucked string	DAMP PICK FDBK MORP
DUST	particle noise	DENS COLR RING MIX
BOW	bowed string	PRES POS DAMP SPD

PAGE	KNOBS 1–4	KNOBS 5–8
MACRO	MODL SAWS SPRD HARM	MORP TUNE FINE LEVL
AMP	ATK DEC SUS REL	CHAN STGS COMB BEND
FILTR	CUT RESO ENV DRIV	TYPE MODE MORP KTRK
F ENV	ATK DEC SUS REL	CURV EVEL VEL WHL
STRIP	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
LFO1	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
LFO2	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Wavetable

`TABL` picks a built-in table, `USER`, a table built from any sample (64 frames, so `POS` scans through the recording), or `KEYS`, a table you define with key frames (below). `MSPD` and `MDPT` move the position by themselves; `ENVP` and `VELP` move it with the envelope and velocity. `INTP` sets how the table is read, from smooth to deliberately coarse. Its last setting, `CLSC`, reads any table the way the first digital wavetable synths did: the scan moves in 64 whole steps, the wave is 128 samples of 8 bits with nothing smoothed in between, and nothing is band-limited to the note, so high notes get a glassy top of tones that are not harmonics. (`STEP` is stepped but clean; `VINT` softens each step.)

After the first nine tables come fourteen more: `VOWL` the vowels A E I O U, `SYNC` an oscillator-sync sweep, `FOLD` a sine driven into a wavefolder, `FM1` and `FM3` FM with a rising index (modulator at 1× and 3× the pitch), `BELL` bar-like partials from a soft to a hard strike, `ORGN` an organ pulled out drawbar by drawbar, `NOIS` noise settling into a tone, `RESO` a resonant filter sweep, `WLSH` square-cornered digital waves, `PRIM` prime-numbered harmonics, `OCTV` octaves only, `PDST` phase distortion from sine to saw, and `COMB` a saw against a sliding copy of itself. All are level-matched and free of aliasing in themselves.

Each of the fourteen has at least one factory preset on `PRESET` (Vowel Lead, Sync Sweep Lead, Fold Bass, FM Table Keys, Glass Pluck, Struck Bar, Drawbar Organ, Noise To Tone, Reso Sweep Bass, Walsh Stepper, Prime Shimmer, Octave Climb, Phase Bass, Comb Phase Pad and others), and three play a table through `CLSC`: Eight Bit Vowels, Glassy Bell and Raw Sync Lead. The same 23 built-in tables can be played by the synth's oscillator slots (kind `WT`).

The `SCAN` page moves the position by itself over a set time, whatever note you play: this is time stretch for a wavetable. `MODE` runs once and holds (`RUN`), loops, goes there and back (`PING`), moves in audible steps (`STEP`) or jumps to random frames (`RAND`). The pass lasts `TIME` seconds, or with `SYNC` on, `LEN` at the tempo; with `SYNC` = `NAT` a USER table plays back at the original recording's length, and `SPD` then slows it or speeds it without changing pitch. The scan starts from `POS`.

`FRMS` chooses how a recording is cut into the USER table. The `A64`, `A128` and `A256` settings align each frame to the sound's pitch period: scanning is far smoother and the table plays in tune with the keys. They work on pitched material; noise and silence are cut the plain way.

Key frames: your own table

The `KEYS` table is made from up to eight single waves, the key frames, each picked out of a sample. You say where each one sits along the table, and Neptune computes the frames in between.

1. On the wavetable track open `WAVE` (`Shift` + bottom pad 5). On this track its first page is the key editor. `KY`, or a top-row pad, chooses one of the eight keys.
2. `SM` gives the key a sample: a factory sample, a file, or a recording made on the `REC` page (a take made while the table is `KEYS` goes to the chosen key). The table switches to `KEYS` and you hear the key at once.
3. `PO` moves the line on the waveform: one cycle is cut there. Where the sample has a pitch the cycle is exactly one period, so the wave plays in tune with the keys whatever the pitch of the recording; in noise or a drum hit a short piece is taken. `ZM` and `SC` zoom and scroll for a finer aim.
4. Do the same for a second and a third key, from the same sample or from others. `PL` sets where each sits along the table; only their order and spacing count, the earliest key is the start of the table and the latest the end.
5. `AL` chooses how the frames between two keys are made. Sweep `POS` (or use the `SCAN` page) and listen.

The same settings are cells on four pages of the track: `K SMP`, `K POS` and `K PLC` hold the sample, position and place of the eight keys, one knob per key, and `KEYS` holds the rest.

`ALGO` BETWEEN TWO KEYS	
`LIN`	A plain crossfade. Harmonics the two keys hold in opposite phase pass through silence.

ALGO	BETWEEN TWO KEYS
STEP	Nothing in between: each key holds until the next takes over.
CLSC	A crossfade stored as an early digital instrument stored waves: 128 samples of 8 bits. With INTP on CLSC and a low KFRM this is the whole classic character.
PHAS	Each harmonic's level and phase move separately, so nothing cancels on the way.
TRNS	Optimal transport: the energy of the wave glides along the harmonics. A formant or a single partial slides from where it is in one key to where it is in the next, instead of fading out in one place and in at another.
VOC	Vocoder style: first the outline of the spectrum (its formants) changes, then the harmonics under it. Half way you hear one key's formants on the other key's harmonics.

KFRM sets the number of frames (8 to 256), BIAS leans the in-between frames towards the earlier or the later key, and WRAP brings the table back to its first key at the end, for a looped scan without a jump. ALGN lines the keys up in phase (leave it on unless you want the raw cuts), NORM evens out the level of the frames, TILT darkens or brightens every key and HARM removes the harmonics above a number. Each change rebuilds the table, so these cells cannot be locked to steps. Key frames are saved with the project; their samples are found again by name.

PAGE	KNOBS 1–4	KNOBS 5–8
WAVE	TABL POS MSPD MDPT	ENVP VELP INTP LEVL
SCAN	MODE TIME SYNC LEN	STEP DIR SPD FRMS
OSC	OCT TUNE FINE BEND	GLID VIBR SPRD SUB
AMP	ATK DEC SUS REL	VEL CHAN STGS COMB
KEYS	ALGO KFRM BIAS WRAP	ALGN NORM TILT HARM
K SMP	SMP1 SMP2 SMP3 SMP4	SMP5 SMP6 SMP7 SMP8
K POS	POS1 POS2 POS3 POS4	POS5 POS6 POS7 POS8
K PLC	PLC1 PLC2 PLC3 PLC4	PLC5 PLC6 PLC7 PLC8
FILTR	CUT RESO ENV DRIV	TYPE MODE MORP KTRK
F ENV	ATK DEC SUS REL	CURV EVEL VEL WHL
STRIP	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
LFO1	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
LFO2	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Tables: SAW · SQR · TRI · SIN · PWM · FORM · HARM · SPEC · DIGI · USER · KEYS · VOWL · SYNC · FOLD · FM1 · FM3 · BELL · ORGN · NOIS · RESO · WLSH · PRIM · OCTV · PDST · COMB.

Wave sequencer

Each note steps through a short sequence of waves. Pitch, shape, level and step time each have their own lane of up to sixteen steps, and the lanes can be different lengths (TLEN PLEN SLEN LLEN), so the combination drifts before repeating. On the lane pages, knob N is step N.

PAGE	KNOBS 1–4	KNOBS 5–8
SEQ	LEN XFAD LOOP RATE	PRNG TUNE FINE LEVL
LANES	TLEN PLEN SLEN LLEN	SYNC SLID SHP* BEND
PIT A	P1 P2 P3 P4	P5 P6 P7 P8
PIT B	P9 P10 P11 P12	P13 P14 P15 P16
SHP A	S1 S2 S3 S4	S5 S6 S7 S8
SHP B	S9 S10 S11 S12	S13 S14 S15 S16
LVL A	L1 L2 L3 L4	L5 L6 L7 L8
LVL B	L9 L10 L11 L12	L13 L14 L15 L16
TIM A	T1 T2 T3 T4	T5 T6 T7 T8
TIM B	T9 T10 T11 T12	T13 T14 T15 T16
AMP	ATK DEC SUS REL	VEL CHAN STGS COMB
FILTR	CUT RESO ENV DRIV	TYPE MODE MORP KTRK
F ENV	ATK DEC SUS REL	CURV EVEL VEL WHL
STRIP	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
LFO1	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
LFO2	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Sampler

Plays a sample from the bank: the built-in loop and hits, WAV or AIFF files you put in *Music / Neptune / Samples*, and anything you resample. `MODE` sets how it plays; `GRAN` switches to granular playback, with its own page. The `SLICE` page cuts the sample into slices, evenly or at its hits, and the `STRCH` page changes its speed without changing its pitch (all in Chapter 11). Edit the markers by eye on the `WAVE` screen.

PAGE	KNOBS 1–4	KNOBS 5–8
SAMPL	SMPL MODE STRT END	LPST LPEN ROOT NORM
GRAIN	SIZE DENS SCAN SPRY	PJIT SPRD SHAP RVRS
SLICE	MODE SLCS SENS GAP	SLIC PLAY FADE SLEN
STRCH	ALGO SPD SYNC LEN	SIZE XFAD JIT SPRD
TONE	PTCH FINE BITS RATE	DRIV LEVL KEY BEND
FILTR	CUT RESO TYPE MODE	VELF ENVF KTRK FDRV
AMP	ATK DEC SUS REL	VEL VOX PAN RAND
STRIP	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
LFO1	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
LFO2	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Drums

Sixteen voices, each a small synthesizer with eight controls. Play and select them on `DRUMS`; edit the selected one there or on `PARAM`. Closed and open hats cut each other off.

PAGE	KNOBS 1–4	KNOBS 5–8
`DR`	TUNE DEC TONE DRIV	SNAP LEVL PAN TYPE
`KICK`	SWEP SWPT SCRV SHAP	KNCK KTON HOLD ACRV
`STRIP`	DST1 AMT1 DST2 AMT2	DST3 AMT3 WVIB VRAT
`LFO1`	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST
`LFO2`	RATE DPTH SHPE TRIG	SYNC DIV FADE DEST

Types: `FMKK` · `KICK` · `MKIK` · `SNAR` · `FMSN` · `CLAP` · `HAT` · `OHAT` · `CYMB` · `TOM` · `RIM` · `SHKR` · `CLAV` · `SUB` · `KSYN` (FM kick, analog kick, modelled kick, noise snare, FM snare, clap, closed hat, open hat, cymbal, tom, rim, shaker, clave, sub-bass, kick machine).

The five sound cells act on every type. `DEC` is the length: a drum is a one-shot and rings for its `DEC` however short the pad press or the step (`SUB`, the bass note, is the exception: it sounds while held and fades over `DEC` afterwards). `TONE` is brightness in the way of each drum (Appendix A lists them). `DRIV` is saturation. `SNAP` is the front edge: 50 leaves the drum as it is, lower softens the onset, higher lifts it and gives the tonal drums a click. Pad pressure (`AT ROUTE`) reaches `DEC`, `TUNE`, `TONE`, `LEVL`, `DRIV` and `PAN` of the hit that is sounding; `SNAP` belongs to the strike, so pressure cannot change it.

The kick machine (KSYN)

The lower left pad of a new kit is a kick machine; set any other pad's `TYPE` to `KSYN` and it becomes one too. A kick machine is a full kick synthesizer with a second page, `KICK`: on `DRUMS` press a knob-page button (the labels become `SW ST SC SH KN KT HD AC`); on `PARAM` it is the page after the pad's own.

- On the pad's first page: `TUNE` is the pitch the drum settles on, `DEC` how long the body rings, `TONE` a low-pass over the whole drum, `DRIV` saturation, `SNAP` how hard the front edge is.
- `SWEP`: how far above `TUNE` the pitch starts, in semitones (0 = no sweep, 72 = six octaves). `SWPT`: how long the fall takes, 2 ms to 1 s. `SCRV`: where in that time the fall happens (negative: nearly all at the start, the classic kick; positive: late).
- `SHAP`: the body's waveform, from a sine through a triangle to a squared-off wave.
- `KNCK`: the level of the knock, a short tick at the very start. `KTON`: its tone, from a dull thud that lasts about 12 ms to a bright tick of about 1.5 ms.
- `HOLD`: time at full level before the decay starts, up to 0.3 s, for long, flat 808-style bodies. `ACRV`: the shape of the decay.

Every one of these can be locked per step and given a pressure range, like any other cell. The other fourteen types are unchanged and ignore the `KICK` page.

Playing the kit

The pads play the kit on `DRUMS` (`Shift` + bottom pad 2); on the other screens the pads do that screen's own job, unless `DRUMS` is pinned. The keys play the kit on every screen while a drum track is selected. From the lowest C, eight keys (C to G) are the bottom pad row from left to right, so the lowest C is the lower left pad; the next eight (G# to D#) are the top row; the last eight (E to B) are the bottom row once more, and the top C is the lower left pad again. Harder

keys are louder. The layout repeats every two octaves, so after one press of an octave button the same rows sit an octave along, and after two the keys are as they were.

> **SEQUENCING DRUMS** Record drum patterns by playing the pads or the keys with Rec on. On STEP you enter one drum at a time: the pads show the hits of the pad last selected on DRUMS. A tap adds that drum to a step or takes it away, and the other drums on the step stay, so any number of the sixteen can hit together. A pad glows dimly, and the step number is underlined, where only other drums hit. Hold a step and press a key to add that key's drum. On the step's TRIG page NOTE and VEL are the selected drum's own; length, timing, condition and locks belong to the step.

MIDI track

Sends notes, pitch bend, per-note pressure, program changes and controllers to outside gear. Its pages send standard controllers, so the same knobs, locks and pressure ranges work on a hardware synth.

PAGE	KNOBS 1–4	KNOBS 5–8
MIDI	CHAN PROG VOL PAN	MOD EXPR CUT RESO
SOUND	ATK DEC REL VRAT	VDPT VDLY RVB CHOR
CC	CC16 CC17 CC18 CC19	CC20 CC21 CC22 CC23

6 Filters

Every filter cell uses the same words: `TYPE` is the model, `MODE` its response, then `CUT` and `RESO`. While you turn any filter knob, the response curve is drawn over the half of the grid you are not using; `FLTR` shows it full-size.

Envelopes have a picture too. On any page with an envelope (`AMP`, `F ENV`, `MENV`, an FM operator's page), turning `ATK`, `DEC`, `SUS`, `REL` or `CURV` draws the envelope's shape over the other half of the grid for a second: the rise, the fall to the sustain level, a held stretch, and the release, with dotted lines where one stage ends and the next begins. `CURV` bends the lines exactly as it bends the sound. Each stage is drawn wider the longer it lasts, but on a compressed scale (a stage ten times longer is a few columns wider, not ten times), so a 1 ms attack and a 10 s release can share the picture.

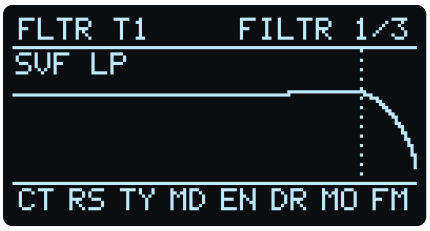
Each stage can be bent on its own. Hold `Shift` and turn `ATK`, `DEC` or `REL`: instead of a fine change of the time, that one stage changes shape. Turned left (down to −100) the move comes early: an attack that bulges upward, a decay or release that drops at once and then levels out. Turned right (up to +100) it comes late: a hollow attack, a decay or release that holds and then falls. 0 is a straight line. While you turn, the cell shows the curve (`ACRV`, `DCRV` or `RCRV`), the name bar says for example "ATK CRV +40", and the picture shows the new shape; a second later the cell shows the time again. This works wherever those three cells belong to an envelope: `MAIN`, `AMP`, `F ENV`, `MENV`, every FM operator, on `FLTR`, and on a held step, where it locks the curve for that step. `SUS` is a level and has no shape, so `Shift` stays fine there; a MIDI track's `ATK`, `DEC` and `REL` are controller values and have no curve either. Where there is a `CURV` cell it still bends all three stages together, and each stage's own bend is added to it. The curves are saved with the sound.

TYPE	WHAT IT IS	MODE	CHARACTER
`SVF`	12 dB state-variable	LP BP HP NTCH	Clean, stable right up to the top; the neutral choice.
`SEM`	Expander-style 12 dB	MRPH BP	Gentle and wide. In MRPH, `MORP` sweeps low-pass through notch to high-pass. Never self-oscillates.
`LADR`	Transistor ladder	LP24 LP18 LP12 LP6 BP HP	The classic thick low-pass. Self-oscillates at full resonance.
`DIOD`	Two-pole with diode limiting	LP HP BP	Rougher, asymmetric resonance.
`COMB`	Tuned feedback comb	POS NEG	Metallic, pitched resonances; cutoff sets the pitch.
`OFF`	Bypassed	–	–

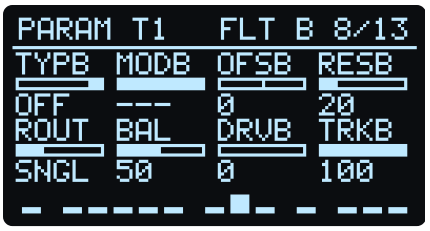
`DRIV` overdrives the filter input, each type in its own way (the SVF soft-clips; on COMB it darkens the feedback). `MORP` acts in one place, type SEM in mode MRPH; anywhere else the knob says `SEM MRPH ONLY`. `ENV` sets how far the filter envelope moves the cutoff, up to seven octaves either way. `KTRK` / `KEY` makes the cutoff follow the keyboard.

The synth has two filters. The `FLT B` page sets the second one: its own type and mode, its cutoff as an offset from filter A (`OFSB`), and `ROUT`: single, series or parallel, with `BAL` to balance them.

FM, macro, wavetable and wave sequencer share one filter design (`FILTR` and `F ENV` pages), open by default so presets sound as they did before you touch it. **The sampler** has its own `FILTR` page. A filter is also available as an insert effect (`FLT`) for any track, drums included.



FLTR



Synth, FLT B

7 Modulation and pressure

LFOs

LFOs run per voice: each note has its own. The synth has up to four, on its MOD and MOD2 pages (choose which with LFO, how many with LFOS). Every other sound engine ends with three pages: STRIP (what the mod strip does, see Strips below), then LFO1 and LFO2.

CELL	WHAT IT DOES
RATE	Speed in Hz. With SYNC on, a tempo division instead (8 bars to 1/32, dotted and triplet included); both values are remembered.
DPTH	Amount; negative inverts.
SHPE	SIN TRI SQR SAW RND SRND RAMP EXPU EXPD — sine, triangle, square, saw, stepped random, smooth random, ramp, and two exponential shapes that only push one way.
TRIG	FREE runs on regardless of notes. KEY restarts with each note. ONE plays one cycle and stops: a one-shot envelope in any shape. HALF plays half a cycle. HOLD takes one value at note-on and keeps it: a different value per note.
DEST	What it moves; see below.
FADE	Fade-in time after the note starts.
PHAS DLY	Synth only: start phase, and a wait before the LFO begins.
WHL VEL PRES	Synth only: the mod strip, velocity, and the note's own pad pressure add to the depth. DPTH 0 with WHL up gives vibrato from the strip alone.
KTRK	Synth only: higher notes speed the LFO up.

Destinations

An LFO can reach nearly everything its engine has a knob for. DEST starts with the engine's basic destinations (pitch, level, pan, cutoff, resonance and the engine's main controls) and goes on with every other continuous cell of the engine's pages, page by page; on the six engines with two LFOs the list ends with the other LFO's rate and depth (LRAT, LDPT). Plain turns go one entry at a time; Func + turn jumps to the start of the next page's group, so the far end of a long list is a few clicks away.

A destination that is a page cell has that cell's name, or the name with a number or an F in front where two pages use the same word (ATK1 is operator 1's attack, FATK the filter envelope's). The LFO moves the cell around the value its knob is set to, in the cell's own scale: at DPTH 100 a 0–100 cell swings by its whole range and a time or frequency cell by half its range in octaves, never past the cell's limits. Turn the knob and the swing moves with it. Every note has its own LFOs, so in a chord each note's cell moves by itself.

ENGINE	DEST OFFERS (AFTER OFF)
Synth	74: PTCH CUT RESO MIX AMP PW XMOD CUTB PAN LVL1 LVL2 LVL3 NOIS FFM PIT2 RAT1 RAT2 RAT3 RAT4 DPT1 DPT2 DPT3 DPT4 · TUN2 DET2 DRIV TUN3 NCOL · 1A 1B 2A 2B 3A 3B · SPRD GLID DRFT DRAT · ENV FDRV MORP RESB BAL DRVB TRKB · FATK FDEC FSUS FREL KEY FVEL PMOD · ATK DEC SUS REL VEL · ATK2 DEC2 SUS2 REL2 · ATK3 DEC3 SUS3 REL3 · ATK4 DEC4 SUS4 REL4 · ATK5 DEC5 SUS5 REL5 EDPT

ENGINE	DEST OFFERS (AFTER OFF)
FM	68: `PTCH` `AMP` `PAN` `CUT` `RESO` `INDX` `FDBK` `DETN` `LVL1` `LVL2` `LVL3` `LVL4` `LVL5` `LVL6` `RAT1` `RAT2` `RAT3` `RAT4` `RAT5` `RAT6` `DET1` `FDB1` `ATK1` `DEC1` `SUS1` `REL1` `DET2` `FDB2` `ATK2` `DEC2` `SUS2` `REL2` `DET3` `FDB3` `ATK3` `DEC3` `SUS3` `REL3` `DET4` `FDB4` `ATK4` `DEC4` `SUS4` `REL4` `DET5` `FDB5` `ATK5` `DEC5` `SUS5` `REL5` `DET6` `FDB6` `ATK6` `DEC6` `SUS6` `REL6` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT`
Macro	26: `PTCH` `AMP` `PAN` `CUT` `RESO` `SAWS` `SPRD` `HARM` `MORP` `LEVL` `ATK` `DEC` `SUS` `REL` `ENV` `DRIV` `FMOR` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT`
Wavetable	33: `PTCH` `AMP` `PAN` `CUT` `RESO` `POS` `SPRD` `SUB` `MSPD` `MDPT` `ENVP` `VELP` `TIME` `SPD` `GLID` `VIBR` `ATK` `DEC` `SUS` `REL` `VEL` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT`
Wave seq	74: `PTCH` `AMP` `PAN` `CUT` `RESO` `RATE` `XFAD` `PRNG` `LEVL` `ATK` `DEC` `SUS` `REL` `VEL` `ENV` `DRIV` `MORP` `KTRK` `FATK` `FDEC` `FSUS` `FREL` `EVEL` `FVEL` `LRAT` `LDPT` `P1` `P2` `P3` `P4` `P5` `P6` `P7` `P8` `P9` `P10` `P11` `P12` `P13` `P14` `P15` `P16` `L1` `L2` `L3` `L4` `L5` `L6` `L7` `L8` `L9` `L10` `L11` `L12` `L13` `L14` `L15` `L16` `T1` `T2` `T3` `T4` `T5` `T6` `T7` `T8` `T9` `T10` `T11` `T12` `T13` `T14` `T15` `T16`
Sampler	41: `PTCH` `AMP` `PAN` `CUT` `RESO` `STRT` `DRIV` `GPOS` `SIZE` `DENS` `SPD` `END` `LPST` `LPEN` `SCAN` `SPRY` `PJIT` `SPRD` `RVRS` `SLIC` `FADE` `SLEN` `SSIZ` `XFAD` `JIT` `SSPR` `BITS` `RATE` `LEVL` `VELF` `ENVF` `KTRK` `FDRV` `ATK` `DEC` `SUS` `REL` `VEL` `RAND` `LRAT` `LDPT`
Drums	17: `PTCH` `DEC` `TONE` `DRIV` `SNAP` `AMP` `PAN` `SWEP` `SWPT` `SCRV` `SHAP` `KNCK` `KTON` `HOLD` `ACRV` `LRAT` `LDPT`

On the synth, `1A` to `3B` are the two `SRC` cells of each oscillator slot; `ATK2` to `REL5` and `EDPT` are the extra envelopes of the `MENV` page. On the wave sequencer, `P1` – `P16`, `L1` – `L16` and `T1` – `T16` are the pitch, level and time of one step.

> **STRUCK VALUES** Some values are read once, when the note is struck, so the LFO sets them per note rather than moving them while the note sounds. Use `FREE` or `HOLD` there; `KEY` would give the same value every time. Drums: `DEC`, `SNAP`, and on a kick-machine pad `SWPT`, `SCRV`, `KNCK`, `HOLD`, `ACRV`. Sampler: `STRT`, `END`, `LPST`, `LPEN`, `SLIC`, `FADE`, `SLEN`, `VELF`, `KTRK`, `ATK`, `DEC`, `SUS`, `REL`, `RAND`; its grain cells `SPRY`, `PJIT`, `SPRD`, `RVRS` are read each time a grain starts.

> **NOT IN THE LISTS** Cells that choose from a list or count something (types, modes, voices, steps), cells that rebuild a sample or a table, whole-sound tuning (`PTCH` is the pitch destination), envelope curves, and `LEVL` where it is the level of the mixed voices (synth, FM, wavetable: use `AMP`).

Extra envelopes

The synth's `MENV` page adds up to four envelopes beyond the amp and filter ones: `ENV` selects, `ENVS` sets how many run, then `ATK DEC` | `SUS REL DPTH DEST`, with destinations amp, cutoff or pitch and then the same long list as the LFOs.

Pad pressure

Each pad sends its own pressure, and it reaches only the note that pad is playing. There are three ways to use it.

1. Routes, per track

`AT ROUTE` gives each track up to four routes: a destination and a depth. A new track starts with pressure on level (drums: decay).

ENGINE	PRESSURE DESTINATIONS
Synth	LEVL CUT VIB PW MIX XMOD
FM	LEVL CUT FM
Macro	LEVL CUT VIB TIMB MORP
Wavetable	LEVL CUT VIBR POS
Wave seq	LEVL CUT
Drums	DEC TUNE TONE LEVL DRIV SNAP PAN (one route for the kit)
Sampler	none (use ranges, below)
MIDI track	sends per-note pressure out

2. Ranges, on any knob

On `PARAM`, `TRK FX` or `FLTR`: turn a knob to a value, then `Func` + ▲ sets that as the high end of a pressure range, or `Func` + ▼ as the low end. The name bar confirms (`RESO HI 80`). The bottom-row pad under that knob turns violet; pressing into it sweeps the knob and lets it return on release. With one end set, pressure moves from the knob's value to that end; with both, from low to high. The same gesture again clears the end.

3. The arp and rolls

Pressure can drive the arp's rate, octave range or gate (`ACFG` page) and the speed of pad rolls (`ROLL`).

Strips

The pitch strip bends the selected track by its engine's `BEND` range (a drum kit: two semitones, the whole kit together) and springs back. It acts on the selected track only: other tracks playing from the sequencer stay in tune.

The mod strip holds its position. It always does something to the selected track, on every engine and every factory preset, and nothing at all when it is at the bottom. What it does is yours to change. On FM, macro, wavetable, wave sequencer, sampler and drums the `STRIP` page (just before `LFO1`) has three routes, `DST1` `AMT1`, `DST2` `AMT2`, `DST3` `AMT3`: each sends the strip to one destination of the list the LFOs use (Func + turn jumps from group to group) by a signed amount in that destination's own scale, so +100 with the strip fully up is where an LFO at full depth would peak. Pitch goes by the amount squared, up to an octave. Routes to the same destination add up, `LRAT` and `LDPT` from the strip act on both LFOs, and an amount of 0 or a destination of `OFF` switches a route off. `WVIB` is vibrato the strip brings in (semitones at the top of the strip) and `VRAT` its speed. The amounts can be locked per step. Until you change the page, the strip does this:

ENGINE	THE MOD STRIP BY DEFAULT
Synth	Goes to every LFO whose `WHL` is set (`MOD2` page), adding to that LFO's depth. While no LFO in use has a `WHL` amount it is vibrato, up to half a semitone. An LFO set to `EXPU`, `ONE`, 40 Hz, `DPTH` 0 passes the strip straight on to its `DEST`: that is how the factory presets and the three `ARCH` sounds open a filter or raise cross-mod from the strip.
FM	`DST1` `INDX` +100: raises the modulation index, up to three times (brighter, harder; from the strip `INDX` counts double). `WVIB` 0.25: a little vibrato. It also moves the cutoff by `WHL` on the `F ENV` page, as on the next three engines.

ENGINE	THE MOD STRIP BY DEFAULT
Macro	`DST1` the fourth macro (`MORP` in most models) +100: sweeps it to its top. `DST2` the first macro +30. `WVIB` 0.25. Cutoff by `WHL` .
Wavetable	`DST1` `POS` +60: scans the table forward from `POS` , up to 60 % of the table. `WVIB` 0.25. Cutoff by `WHL` .
Wave sequencer	`DST1` `RATE` +50: runs the sequence faster, twice as fast at the top (so the top of the strip is in time again). Cutoff by `WHL` .
Sampler	Wears the sound down: `DST1` `DRIV` +70, `DST2` `RATE` +30 (sample-rate reduction, `TONE` page), `DST3` `SPRY` +50 (scatters the grains in `GRAN` mode). Whatever the strip adds to `DRIV` is mostly taken out of the level again, so the tone changes more than the loudness.
Drums	Opens the whole kit up: `DST1` `DEC` +79, every drum rings up to three times as long (from its next hit), and `DST2` `DRIV` +50, driven harder. The routes belong to the kit: every pad follows them.
MIDI	Sends controller 1 to the instrument outside.

The strip moves the selected track only, and every track keeps the value the strip last gave it. Set one track's strip, select another and move the strip there: the first stays as you left it. Selecting a track changes nothing until you touch the strip, and a preset you load on a track is given that track's value again. While a track's arp is on, the strip can drive one arp macro instead (`STRP` on the arp's `VARY` page); the track's sound then keeps whatever the strip last gave it.

8 Effects

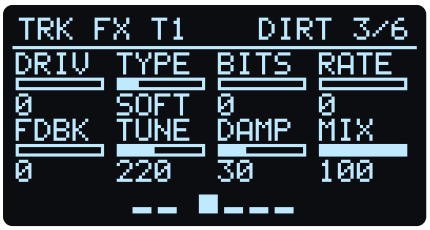
Signal path

Engine → `DIRT` → `TALK` → `MODFX` → insert slots → mixer (volume, pan) → master EQ, compressor, limiter. Each track also sends to two shared buses. All of a track's effect pages are on `TRK FX`.

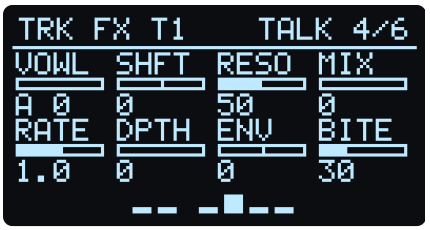
The three character blocks

On every track, always available, bypassed while at zero.

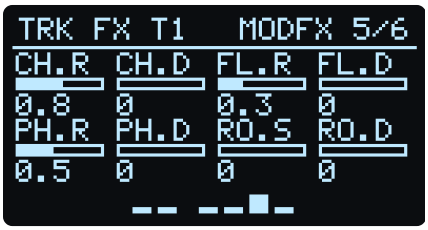
- `DIRT`: `DRIV TYPE BITS RATE` | `FDBK TUNE DAMP MIX`. Distortion with four curves, bit and sample-rate reduction, and a tuned feedback path that turns it into a resonator.
- `TALK`: `VOWL SHFT RESO MIX` | `RATE DPTH ENV BITE`. A vowel filter. The vowel can be moved by hand, by its own LFO, or by how hard you play.
- `MODFX`: chorus, flanger, phaser and rotary speaker, a rate and a depth each (`CH.R CH.D` | `FL.R FL.D` | `PH.R PH.D` | `RO.S RO.D`).



DIRT



TALK



MODFX

Insert effects

The `FX` page shows one insert slot: `SLOT` chooses which, `TYPE` the effect, `WET` how much of it, and knobs 5 to 8 are the effect's own four controls.

> **CAREFUL** Turning `TYPE` replaces the effect and its four values at once, without the four-click confirmation. Undo (`Func` + `Shift` + `Play`) brings the effect back as it was; a scroll through the list counts as one step.

TYPE	EFFECT	KNOBS 5-8
`CHR`	Chorus	`RATE` `DPTH` `MIX` `VOIC`
`DRV`	Drive	`DRIV` `TONE` `MIX` `ALGO`
`FLG`	Flanger	`RATE` `DPTH` `FDBK` `MIX`
`PHS`	Phaser	`RATE` `DPTH` `FDBK` `STGS`
`TRM`	Tremolo	`RATE` `DPTH` `SHPE` `STER`
`DLY`	Delay	`TIME` `FDBK` `MIX` `MODE`
`RVB`	Reverb	`DCAY` `DAMP` `SIZE` `MIX`
`EQ`	Three-band EQ	`LOW` `MID` `MFRQ` `HIGH`
`CMP`	Compressor	`THR` `RATI` `ATK` `MAKE`

TYPE	EFFECT	KNOBS 5–8
LMT	Limiter	CEIL REL LOOK
FLT	Filter	CUT RESO TYPE MODE
ENS	Ensemble	RATE DPTH WDTH MIX
PAN	Auto-pan	RATE DPTH SHPE PHAS
ROT	Rotary speaker	SPD HORN DRUM ACCL
WSH	Waveshaper	DRIV CURV TONE ASYM
RED	Rate reducer	RATE BITS FILT MIX
HALL	Hall reverb	DCAY SIZE DAMP PDLY
SPRG	Spring reverb	DCAY DISP TENS DAMP
SHIM	Shimmer reverb	DCAY SHIM MODE DAMP
RVSE	Reverse delay	TIME FDBK HCUT MIX
TRNS	Transient shaper	ATK SUS SPD OUT
RING	Ring modulator	FREQ RATE DPTH MIX
VIB	Vibrato	RATE DPTH SPRD FADE
WAH	Wah	FREQ RESO SENS RATE
AMP	Amp simulator	GAIN TONE CAB LEVL
GEQ	Four-band graphic EQ	LOW LMID HMID HIGH
TAPE	Tape delay	TIME FDBK AGE MIX
LCR	Left-centre-right delay	TIME FDBK SPRD MIX
MTAP	Multi-tap delay	TIME TAPS FDBK MIX
DUCK	Sidechain compressor	SRC THRS RATI RELS

Tempo sync

An insert that has a TIME (the delays) or a RATE (chorus, flanger, phaser, tremolo, auto-pan, ensemble, ring modulator, vibrato, wah) shows SYNC in cell 3 in place of ON. With SYNC on, that cell picks one of 22 divisions from 1/128 to 16 bars, triplets and dotted included, and follows the tempo. The free value is kept for when you switch sync off. A division too long for the effect is halved until it fits. To take a synced effect out, turn WET to 0.

Sends and master

SND1 and SND2 on each track feed two buses shared by all tracks: a reverb and a delay to begin with. BUS1 and BUS2 on the TRACK page change what each bus carries, from the same list of effect types.

The master EQ, compressor and limiter are on SETTINGS → MASTR; master volume is MVOL on SETUP.

Sidechain ducking

The DUCK insert turns its track down when another track plays: SRC chooses which (SELF makes it an ordinary compressor), THRS and RATI how hard, RELS how fast it comes back.

9 The sequencer

Patterns and tracks

A project has 16 patterns. Each pattern holds the steps of every track; each track has its own length (`LN` , 1 to 1024 steps) and speed (`SP` : quarter, half, normal, double, quadruple). A track has one sound for all patterns.

Hold [>] and press a pad to queue a pattern. While playing, the change happens at the end of the loop; stopped, at once.

Steps

On `STEP` , a tap adds or removes a step and a held step can be edited. Hold a step and:

- press keys to set its note or chord;
- keep holding: the display changes to the track's `MAIN` page showing that step's values, and any knob you turn is locked on that step;
- use the knob-page buttons to reach the other sound pages, and, one press back from `MAIN` , its `TRIG` page: `NOTE` , `VEL` , `LEN` (length), `MICR` (micro-timing, up to half a step early or late), and the condition cells.

Parameter locks

A lock is a knob value that applies to one step only. Hold a step on `STEP` , `PARAM` , `TRK FX` or `FLTR` and turn a knob. While the step is held the cells show that step's values and the name bar shows its number (`S5`). After the step, the knob returns to its normal value. Hold several steps and the same turn locks all of them to the value shown. Engine controls, mixer levels and sends, effect blocks and inserts, and the arp's gate, velocity and ratchet can all be locked.

Remove one lock: hold the step (or several), hold [>], turn the knob. Remove all of a step's locks: `LCKS` on its `TRIG` page, anticlockwise (an undo step).

Pressure into locks. On a step with recorded pad pressure, `>LCK` on its `PRES` page (four clicks) writes the swell as locks: on every step the note spans, each cell the track's pressure routes point at is locked to where the average pressure over that step would hold it. A route's depth is how far full pressure moves the cell from its own value toward its top (toward its bottom when the depth is negative). Steps under the note that had no trig become trigs that carry only locks, with no note of their own. Chords on a synth track share the cells, so their pressures are averaged; on a drum track each pad's own cell is locked. A route with no cell to lock (vibrato; `FM` on the FM synth; `TIMB` on the macro oscillator) is left out and the name bar says so (`2 OF 3 ROUTES`); if no route can be locked, nothing is changed (`NO ROUTE`). The recorded pressure is then removed: from here on the swell is edited like any other lock.

Trig conditions

COND	THE STEP PLAYS...
`ALL`	every time
`PROB`	with the chance set by `PROB`
`A:B`	on pass `A` of every `B` passes of the track (1:2 = every other time, 4:4 = every fourth)
`FILL` / `!FIL`	only while fill is held ([`Shift`] + [`Rec`]) / only while it is not
`1ST` / `!1ST`	only on the first pass after starting / on every pass but the first

COND	THE STEP PLAYS...
PRE / !PRE	only if the previous conditional step played / only if it did not

Recording

Start. Hold Rec and press Play: recording goes on and the sequencer starts from the top. If it is already playing it keeps playing and recording simply switches on. Rec alone switches recording on or off without touching the transport (it acts when you let the button go), so Rec, then Play works too.

Metronome. While recording runs a click marks every beat, higher and louder on the first beat of each bar. CLCK on SETTINGS → CLICK chooses when it sounds: OFF, REC (only while recording runs: the setting it starts with) or PLAY (whenever the sequencer plays). CLVL sets its level; each turn of that knob plays one click, so you can set it before you start. The click follows the tempo, stops with the sequencer, and is only for your ears: it does not pass through the master effects or MVOL, and it is never recorded when you resample a track or the master.

Count-in. Set CNT on the same page to 1BAR or 2BAR and a recording starts with that many bars of clicks: Rec + Play (or Play with recording already on) shows COUNT IN 1 BAR, the metronome counts four beats to the bar – also when CLCK is off – and the sequencer and the recording start together on the beat after. During the count-in nothing plays, nothing you play is recorded, and a queued pattern waits. Play during the count-in calls it off. The count-in is only for a recording that starts the sequencer: if the sequencer is already playing, recording switches on at once, and Play without recording always starts at once.

While recording the name bar shows REC FREE or REC QUANT on every screen (a single R or Q when room is short) and a red light runs across the pads, one pad per step of the selected track. Pads that play notes keep playing them under the light; on STEP the playhead is the light, and recorded notes appear on the pads as you play them. Keys are written into the selected track, pad hits (each repeat of a roll included) into the track the pads play, arp output into the arp's track. Notes played together land on one step. A later pass adds to what is there; playing on a step that already holds notes replaces them.

Pressure is recorded with pad notes. What you do with a pad after striking it (press harder, ease off) is recorded with that note, for as long as you hold the pad, and the note is given the length you held it. On playback each note gets its own pressure back, through the track's pressure routes (AT ROUTE) as they are set at that moment, or as polyphonic pressure on a MIDI track: two pads held together keep their own swells. The pressure follows the tempo, the track's speed, micro-timing and swing with its note, and is let go when the note ends, when the track is muted and when the sequencer stops. A pad played by hand over a recorded note of the same pitch has the say while it is held. A note you do not press into stores nothing. A pad that is rolling (ROLL) uses its pressure for the rate of the roll, so that pressure is not recorded; with a roll threshold set, a pad held below the threshold is an ordinary note and its pressure is recorded. Keys have no pressure. A note can be at most one pass of its track long; a pad held longer is recorded up to there. Recording over a step replaces its notes and their pressure. See PRES on a held step's pages to scale, convert or remove what was recorded.

Free or quantized. Press Rec + Play again while recording to change between the two; the choice is kept for the next recording. REC FREE writes each note on the nearest step and keeps its distance from the grid as micro-timing, so it plays back as you played it; QN on STEP pulls playback toward the grid by any amount and can be turned back down later. REC QUANT writes each note exactly on the nearest step.

Stop. Play stops the sequencer and ends recording; Play again plays back what you recorded. Rec alone ends recording and leaves the sequencer playing. Either way, if you recorded anything, STEP opens with the track you recorded last selected, on the step page that holds its first recorded step: hold a step and the display shows that track's MAIN page for it; turn a knob to lock a parameter on that step. (If pads are pinned, the pads stay the pinned ones until you unpin them.)

Copy, keep, fill, tempo

- **Copy / clear / paste**: Func + Rec / Play / >, acting on the held step, the track (STEP), the pattern (TRACKS , clear and paste held for half a second), the sound (PARAM , FLTR) or the effects (TRK FX).
- **Undo / redo**: Func + Shift + Play / Rec, fifteen steps back. A recording is one step.
- **Keep and reload**: > + Rec keeps a temporary copy of the pattern; > + Play brings it back, at once (it is meant to be played in time) and as an undo step. The copy lives until you quit.
- **Fill**: hold Shift + Rec; steps with the FILL condition play while you hold.
- **Tempo**: BPM on SETTINGS , knob 8 on STEP , or tap Func + Settings .
- **Swing**: SWNG on SETTINGS → SETUP .
- **Mute and solo**: bottom pads on TRACKS , or hold ▲ / ▼ and press a bottom pad from anywhere. A muted track keeps its place and comes back in time.

Song mode

SONG is an ordered list of rows, each a pattern, a repeat count and a transposition for pitched tracks. Switch SG on and Play follows the list; LP loops it.

Clock

CLK on SETTINGS sets MIDI clock: INT (Neptune is the clock), EXT (follow incoming clock) or AUTO . LINK is listed for the future and does nothing yet.

10 Arpeggiator and rolls

The arpeggiator

Every track has its own. `Arp` opens the screen; pressed again there, it switches the selected track's arp on. `Shift` + `Arp` latches it so it carries on after you let go (with the arp off, the same gesture latches the keys themselves). Arps follow the transport, the scale lock and the tempo.

PAGE	KNOBS 1–4	KNOBS 5–8	FOR
`ARP`	MODE RATE OCT GATE	RPT SWNG GLID LTCH	The basics
`PLAY`	STRM TIME DIR VEL	RTCH RDEC HUMN FREZ	Strumming, ratchets, humanize, freeze
`VARY`	SPCE DICE DENS SCAT	RCHN GVAR STRP VAR	Variation macros
`MUTA`	MUTA INTV MODE SPCE	EVOL ANC >PAT	Mutation over time
`ACFG`	DEST DPTH REC LOCK	TAP RHYT	Pressure, recording, tapped rhythm
`ROLL`	ROLL MODE BASE MAX	DPTH GATE P>V DCAY	Pad rolls
`ROL2`	SWNG GLID THRS LTCH	HUMN ACNT PROB RPTS	Pad rolls, continued

Modes: Up; Down; Up-down; Random; As played; Chord repeat; Walk; Play order; Converge; Diverge; Poly arp (all notes, accented); Random with octave jumps.

Variation

The `VARY` macros change what the arp plays without changing what you hold: `DENS` thins it out, `SCAT` throws notes up and down octaves, `RCHN` adds ratchets, `GVAR` varies note lengths, `SPCE` sets how bold all of that is and `DICE` re-rolls it. `STRP` hands one of them to the mod strip.

Mutation

With `MUTA` on, the pattern changes itself every `INTV` bars. `MODE` decides whether each change starts from the ancestor again (`ANC`) or builds on the last (`CUM`). `EVOL` shows the generation: turn it up to mutate now, down to go back to the ancestor. `>PAT` (four clicks) writes what the arp is playing into the track's pattern as ordinary steps. Going back to the ancestor and `>PAT` can both be undone.

Tapped rhythm

Switch `TAP` on and tap a rhythm on any pad (they light amber). Switch it off, switch `RHYT` on, and the arp plays its notes in your rhythm.

Locking the arp

`TRK FX` → `ARPLK` holds the arp's gate, velocity and ratchet as lockable cells, so steps of the pattern can change how the arp behaves as it passes.

Pad rolls

With `ROLL` on, any pad that plays a note (`DRUMS` , `SCALE` , `AT ROUTE` , `PRESET`) repeats while held. Pressure raises the rate from `BASE` to `MAX` : tempo divisions in `QUANT` mode, free Hz in `FREE` mode. `P>V` lets pressure set the loudness of each repeat; `DCAY` fades them.

`ROL2` refines it: swing, a glide between rates, a pressure threshold so light touches play single hits (`THRS`), latch, humanize, an accent every Nth hit, a probability per repeat and a limit on the number of repeats. Rolls are recorded hit by hit.

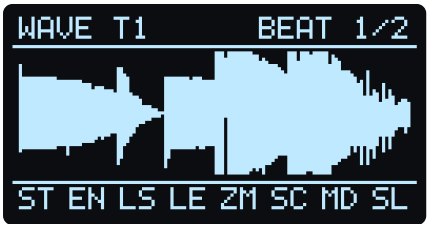
11 Sampling, slicing and stretch

Samples

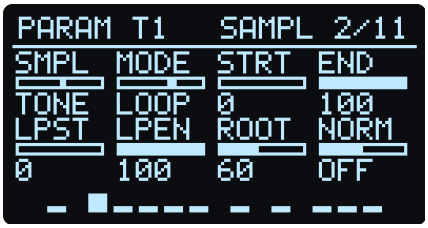
The sampler's `SMPL` cell chooses from the bank: a built-in drum loop, eight hits and a tone, any WAV or AIFF file in *Music / Neptune / Samples* (read at start-up), eight drum breaks (below), and your resampled takes. Set start, end and loop on the `SAMPL` page or by eye on `WAVE`; `Func` + turn is coarse and `Shift` + turn fine, and zooming (`ZM`) makes the markers finer still.

Play modes: `FWD` , `REV` , `1SHT` (plays to the end regardless of the key), `GATE` (stops on release), `LOOP` , `PING` (loops back and forth), `GRAN` .

`ROOT` is the key at which the sample plays at its recorded pitch; `NORM` brings a quiet sample up to full level.



WAVE



Sampler, SAMPL

Drum breaks

Eight drum breaks come with the app, after your own files in the `SMPL` list. The name is the tempo the break was played at: B080, B090, B105, B120, B157 and B160 are recordings of drummers, released by them into the public domain (CC0; names and sources are in *docs/sample-credits.md*), and B100 and B170 are made by the app's own drum voices. Each is one, two or four whole bars. Choosing a break sets `LEN` on the `STRCH` page to its length, so turning `SYNC` on is all it takes to play it in time with the project.

Every break has two sampler presets on the `PRESET` screen (category PERC):

- **Break 90 Loop**: hold a key and the break goes round in time with the tempo, whatever tempo it was played at. It is cut at its drum hits (`MODE` = `TRNS`) and played with `ALGO` = `SLCE` , so each hit sounds as recorded and only its place in time moves; other keys play it higher or lower without changing the timing. One voice: a new key starts the break again.
- **Break 90 Slices**: the sixteen strongest hits, one on each key going up from C4 (`PLAY` = `KEYS` , play mode `1SHT`), at their recorded speed and pitch. The pads of `PRESET` play sixteen keys in a row from the C of the track's octave: at C4 that is one slice per pad. Record them into steps to build your own beat from the break.

From there everything on the sampler's pages applies: `ALGO` = `TAPE` for the repitched sound, `PLAY` = `RAND` or `NEXT` on the slices, `SLCS` for fewer or more of them.

Slicing

The `SLICE` page cuts the sample into pieces you can play separately.

- `MODE` = `GRID` : `SLCS` equal slices. Right for a loop that is already in time.
- `MODE` = `TRNS` : one slice per hit, found by listening for the sudden rise at the start of each sound. `SENS` sets how quiet a hit may be and still count; `GAP` sets the shortest time between two hits. `SLCS` shows how many were

found, and turning it down keeps only the strongest. Every slice starts on a zero crossing, so slices do not click. Hits closer together than about a fiftieth of a second are heard as one.

`PLAY` decides which slice a note plays:

- `ONE` : the slice chosen with `SLIC` (or pad N on `WAVE`). Lock `SLIC` per step to re-order a loop.
- `KEYS` : each key plays its own slice, going up from `ROOT` . The loop is laid out across the keyboard.
- `NEXT` : every new note plays the next slice, so a row of identical steps walks through the loop.
- `RAND` : a random slice per note.

`FADE` softens the end of each slice and `SLEN` shortens them all for a gated feel. The `WAVE` screen draws a tick at each slice and lights the pad of the slice that is sounding.

> **NOT LOCKABLE** `SMPL` , `NORM` and the cells that re-cut the sample (`MODE` , `SLCS` , `SENS` , `GAP`) cannot be locked to a step or driven by pressure: each change re-analyses the whole sample. `SLIC` , `PLAY` , `FADE` and `SLEN` can.



Sampler, SLICE

Time stretch

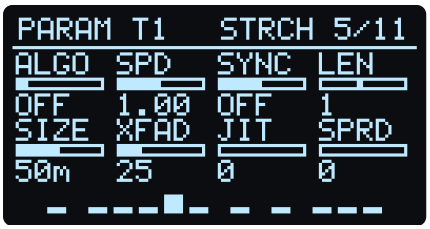
The `STRCH` page changes how long the sample lasts without (except in `TAPE`) changing its pitch. The key still sets the pitch; start, end and loop points, play modes, filter, envelope and LFOs all work as usual. Each `ALGO` has a sound of its own, and that is the point of having several.

ALGO	HOW IT WORKS	SOUNDS LIKE
`TAPE`	Plays faster or slower, like tape or vinyl.	Pitch drops as it slows. With SYNC, the classic repitched loop.
`CYCL`	Repeats short cycles of the sound, crossfading from one to the next.	The metallic, buzzing stretch of early hardware samplers. SIZE tunes the buzz.
`GRAN`	Overlapping grains from the granular engine.	Smooth and slightly diffuse. Good on pads, vocals, textures.
`SLCE`	Does not stretch anything: each slice plays untouched and is started when its turn comes.	Tight and punchy on drums. Slowing down leaves gaps between hits; speeding up cuts them short.
`SMER`	Very long grains from random nearby positions, at a sixteenth of the shown speed.	A frozen wash. A second of sound becomes a slowly moving chord.
`WSOL`	Like CYCL, but each cycle is lined up with the last by matching their waveforms.	The cleanest of the set on single notes and voices.

`SPD` sets the speed from a quarter to four times. With `SYNC` on, the sample between its start and end markers is made to last exactly `LEN` (a sixteenth of a bar to 32 bars) at the current tempo, and follows the tempo if you change it: a loop recorded at any speed locks to the sequencer. `SIZE` , `XFAD` and `JIT` shape the grain of the stretch, and `SPRD` widens `GRAN` and `SMER` .

The LFO destination `SPD` moves the speed by up to two octaves either way: tape-stop dives, or a stretch that breathes.

In `GRAN` play mode the grain page is in charge and `ALGO` is ignored, but `SYNC` still works: `SCAN` at 1.00 then means the region lasts `LEN`.



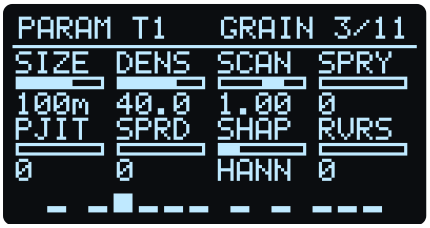
Sampler, STRCH

Granular

In `GRAN` mode the sampler plays the sample as a cloud of short grains. Pitch comes from the key; time is separate. The `GRAIN` page:

- `SIZE` and `DENS`: how long each grain is and how many start per second. Long and dense is smooth; short and sparse is granular in the gritty sense.
- `SCAN`: how fast the read point travels through the sample. 1× is normal speed, 0 freezes on the spot, negative runs backwards. With `SCAN` at 0, `STRT` is the position: turn it, lock it, or put an LFO on `GPOS`.
- `SPRY`: scatters grain positions and timing. `PJIT`: random pitch per grain.
- `SPRD` stereo spread, `SHAP` the grain's envelope, `RVRS` the share of grains played backwards.

The `WAVE` screen shows each grain as a tick while it sounds.



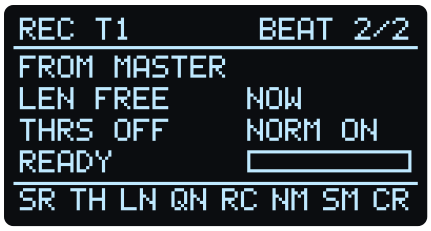
Sampler, GRAIN

Resampling

The `REC` page of `WAVE` records Neptune's own sound into a new sample.

1. `SR`: the source, the master output or one track after its effects.
2. `LN`: free length, or a set length from one beat to eight bars. `QN`: start now or on the next bar. `TH`: optionally wait for the source to pass a level.
3. `RC`: two clicks clockwise within a second arms and starts; one click anticlockwise stops. The top pads are red while recording.
4. The take is saved as *R001.wav*, *R002.wav*… in *Music / Neptune / Samples* and given to the selected track: as its sample on a sampler track, as its `USER` table on a wavetable track (or, while that track's table is `KEYS`, as the sample of the key chosen on the `WAVE` page).

`NM` normalises takes, `SM` picks a different sample for the track, and `CR` (four clicks) crops the sample to its markers.



WAVE, REC page

12 Projects, presets and the rest

Projects

A project is everything: tempo, all 16 patterns with their steps and locks, every track's engine and sound, effects, sends, master section, song, scale pages, pressure routes and ranges, and arp settings. There are 32 slots (*Music / Neptune / Projects*, P01 to P32).

`SETTINGS` → `PROJ` : `SLOT` chooses, `LOAD` , `SAVE` , `SVAS` (save as, to the chosen slot) and `NEW` each need four clicks and answer OK or FAIL. `AUTO` keeps saving to the current slot as you work. The app reopens the last project when it starts.

Not saved: an arp's mutation state, the temporary pattern copy, the clipboards, and the keyboard split.

Presets

A preset is one track's sound. The built-in ones are marked * in the list, among them a loop and a sliced preset for each of the sampler's drum breaks (chapter "Sampling, slicing and stretch"); yours are saved from `PRESET` with `SV` into *Music / Neptune / Presets*. Categories: BASS, PAD, LEAD, KEYS, FX, PERC, KIT, INIT.

Naming: `SV` saves at once as User 1, User 2 and so on, so nothing waits for the Mac. The Mac then shows a box asking for a name: type up to 15 characters on its keyboard and press Return, or press Escape to keep the User name. A name already in use is refused and the name bar says so.

Keyboard split

`Shift` + `Oct−` drops the lowest twelve keys by one further octave, then two, then three, then back to normal; `Shift` + `Oct+` steps the other way. The name bar shows `SPLIT −1 OCT` . Plain `Oct−` / `Oct+` still shift the whole keyboard.

Latency and headroom

Neptune adds about 1.3 ms of its own between a key and its sound. What you hear beyond that is the audio output: an interface or wired headphones add a few milliseconds, the Mac's built-in speakers about 19 ms.

The heaviest test session (eight tracks, eight notes each, every effect block plus delay and reverb on every track) uses about a quarter of one processor core on an M3 Pro. You are unlikely to run out.

The display redraws about twelve times a second, which is as fast as the Launchkey accepts pictures.

If something seems wrong

WHAT YOU SEE	WHAT TO CHECK
No sound	`MVOL` on `SETTINGS` ; the track's `VOL` ; whether it is muted (red pad on `TRACKS`) or another track is soloed (blue).
Pads do not play the drums	Only `DRUMS` has drum pads: `Shift` + bottom pad 2. The keys play the selected drum track on every screen.
A knob does not seem to change the sound	A step lock or a pressure range may be overriding it. Orange pads on `PARAM` have locks; a violet pad has a range.

WHAT YOU SEE	WHAT TO CHECK
Pads play notes on a screen where they should not	They are pinned (`PIN` in the name bar). On `DRUMS` or `SCALE`, hold `Shift` and hold that screen's bottom pad for half a second.
Notes hang or the arp will not stop	Latch is on (`LATCH` in the name bar): `Shift` + `Arp`.
A sound or an effect rings on and will not stop	`Shift` + `Play` stops the sequencer, lets every note go and cuts every effect's tail.
Low keys are in the wrong octave	A keyboard split is on: `Shift` + `Oct+` until the `SPLIT` message says off.
Copy or paste did nothing	They act only on `STEP`, `TRACKS`, `PARAM`, `FLTR` and `TRK FX`, or on a held step. On `TRACKS` clear and paste need the button held for half a second.
A synced delay is shorter than the division says	The division was longer than the delay can hold and has been halved to fit.
The display is blank after plugging in	Quit and restart the app with the Launchkey already connected.

Appendix A Short names

Every short name the display can show on a knob, in alphabetical order. **Belongs to** names the engine, effect or screen and, after the colon, the page. Names that mean different things in different places have one row per meaning. Values shown as 0–100 are the display's scale.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
A	A of A:B	STEP: TRIG	1–8	With COND = A:B the step plays on pass A of every B passes.
ACCL	Acceleration	Insert effect: ROT	0–100	How quickly the rotors change speed.
ACNT	Accent	ARP: ROL2	every Nth	Accents every Nth repeat.
ACRV	Kick decay curve	Drums: KICK	−100…+100	Kick machine: shape of the decay; negative falls fast then lingers.
AGE	Tape age	Insert effect: TAPE	0–100	Wow, flutter and dulling of the tape.
AL	Algorithm	WAVE: WAVE (wavetable track)	LIN … VOC	How the frames between two keys are made (ALGO).
ALGN	Align	Wavetable: KEYS	OFF ON	On: every key is turned in time until its fundamental starts like a sine, so waves cut from different places do not cancel each other. Off: as they were cut.
ALGO	Algorithm	FM: FMMIX	6>1 2+3 3x2 2x3 1>3 ADD 2+4 4+2	How the six operators are wired (8 algorithms, from one chain of six to six parallel carriers).
ALGO	Drive algorithm	Insert effect: DRV	SOFT HARD FOLD CRSH	SOFT, HARD, FOLD or CRSH (bit-crush).
ALGO	Stretch algorithm	Sampler: STRCH	OFF TAPE CYCL GRAN SLCE SMER WSOL	OFF, TAPE (pitch follows speed), CYCL (vintage cyclic stretch, metallic), GRAN (smooth granular), SLCE (slices keep their sound, only their timing moves), SMER (extreme smear), WSOL (clean stretch).
ALGO	Key-frame algorithm	Wavetable: KEYS	LIN STEP CLSC PHAS TRNS VOC	How the frames between two keys of the KEYS table are made. LIN: a crossfade. STEP: none, the nearer key. CLSC: a crossfade kept as 128 samples of 8 bits. PHAS: level and phase of each harmonic separately, so nothing cancels on the way. TRNS: the energy glides along the harmonics (optimal transport), so a formant slides instead of fading. VOC: first the outline of the spectrum changes, then its harmonics; half way it is one key's formants on the other's harmonics.
AMP	Level	all engines: LFO DEST	destination	The voice's level, from silence to double: tremolo.
AMT1	Strip route 1 amount	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	−100…+100	STRIP page: how far the strip, fully up, moves the destination of route 1, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off.
AMT2	Strip route 2 amount	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	−100…+100	STRIP page: how far the strip, fully up, moves the destination of route 2, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
AMT3	Strip route 3 amount	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	−100…+100	STRIP page: how far the strip, fully up, moves the destination of route 3, in that destination's own scale: +100 is what an LFO at full depth reaches at its peak. Negative moves it the other way; 0 switches the route off.
ANC	Ancestor	ARP: MUTA	ORIG / KEPT	What mutations return to: the original pattern or the last kept one.
ARCH	Architecture	Synth: VOICE	INIT MONO HYBR POLY	Loads a starting voice architecture into the same pages: INIT, MONO, HYBR (hybrid) or POLY.
ASYM	Asymmetry	Insert effect: WSH	0–100	Bends the curve to one side for even harmonics.
ATK	Attack	Insert effect: CMP	0.1–100 ms	How fast the compressor reacts, in milliseconds.
ATK	Attack gain	Insert effect: TRNS	−100…+100	Boosts or softens the start of each hit.
ATK	Attack (CC 73)	MIDI track: SOUND	0–127	Sends the attack-time controller.
ATK	Attack	Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP	varies	Time the envelope takes to rise to full level.
ATKn	Envelope attack	Synth, FM: LFO DEST	destination	FM, ATK1 to ATK6: the attack of operator 1 to 6. Synth, ATK2 to ATK5: the attack of an extra envelope (MENV).
AU	Audition	PRESET: bar	OFF / ON	On (the default): the highlighted preset takes over the track's sound for listening only; the previous sound returns on leaving unless LD is used. Off: the list scrolls without changing the sound.
AUTO	Autosave	SETTINGS: PROJ	OFF / ON	Keeps saving the full state to the current slot.
B	B of A:B	STEP: TRIG	1–8	The cycle length for A:B.
BAL	Filter balance	Synth: FLT B	0–100	Balance between filter A and filter B.
BASE	Base rate	ARP: ROLL	division or Hz	Repeat rate at the lightest pressure.
BEND	Bend range	Synth: OSC3; FM: RATIO; Macro: AMP; Wavetable: OSC; Wave seq: LANES; Sampler: TONE	0–24 semitones	How far the pitch strip bends the sound, in semitones.
BIAS	Bias (wavefolder model)	Macro: MACRO	0–100	Third macro of the FOLD model: pushes the wave off-centre into the folder, either way of 50, which brings in the even harmonics.
BIAS	Key bias	Wavetable: KEYS	−100…+100	Leans the frames between two keys towards the earlier key (down) or the later one (up).
BITE	Bite	TRK FX: TALK	0–100	Adds harmonics before the formants so quiet or pure sounds still talk.
BITS	Bit reduction	Sampler: TONE; TRK FX: DIRT; Insert effect: RED	0–100	Lowers the bit depth for digital grit.
BPM	Tempo	SETTINGS: SETUP	20–300	Tempo. Shift = 0.1 steps, Func = steps of 10.
BUS1	Bus 1 effect	TRK FX: TRACK	30 types	Which effect send bus 1 carries. Shared by all tracks.
BUS2	Bus 2 effect	TRK FX: TRACK	30 types	Which effect send bus 2 carries. Shared by all tracks.
CAB	Cabinet	Insert effect: AMP	1500 Hz–10 kHz	Cutoff of the speaker-cabinet filter.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
CC16–CC23	General controllers 16 to 23	MIDI track: CC	0–127	CC page: eight free controllers for whatever the external instrument maps them to.
CEIL	Ceiling	Insert effect: LMT	−3…0 dB	Highest level the limiter lets through.
CEIL	Master limiter ceiling	SETTINGS: MASTR	−3–0 dB	Highest level the master limiter lets through.
CHAN	MIDI channel	MIDI track: MIDI	1–16	Channel the track sends on.
CHAN	Envelope chain	Macro, Wavetable, Wave seq: AMP	OFF SER PAR	OFF, SER (stages follow each other) or PAR (stages run together).
CH.D	Chorus depth	TRK FX: MODFX	0–100	Amount of chorus; 0 bypasses it.
CHEB	Chebyshev (wavefolder model)	Macro: MACRO	0–100	Fourth macro of the FOLD model: harmonic-selective shaping.
CHOR	Chorus send (CC 93)	MIDI track: SOUND	0–127	Sends the chorus-level controller.
CH.R	Chorus rate	TRK FX: MODFX	0.1–5 Hz	Speed of the block's chorus.
CLCK	Metronome click	SETTINGS: CLICK	OFF REC PLAY	When the metronome clicks: never, while recording runs (the default), or whenever the sequencer plays. One click a beat, the first beat of each bar higher and louder. Kept between sessions, not in the project.
CLK	Clock source	SETTINGS: SETUP	INT EXT AUTO LINK	MIDI clock: internal, external, or automatic. LINK is listed but not available yet.
CLR	Clear pressure	STEP: PRES	four clicks	Removes the pad pressure recorded with the step's notes. The notes stay.
CLVL	Click level	SETTINGS: CLICK	0–100	Loudness of the click. Every turn plays one click, so it can be set with the sequencer stopped. The click is added after the master effects and MVOL: they do not change it, and it is never recorded into a resample.
CNT	Count-in	SETTINGS: CLICK	OFF 1BAR 2BAR	Bars the metronome counts in before a recording starts. The count-in always clicks, also with CLCK off. Nothing plays or is recorded until it ends.
COLR	Colour (particle noise model)	Macro: MACRO	0–100	Second macro of the DUST model: tone of the particles.
COMB	Chain combine	Macro, Wavetable, Wave seq: AMP	SUM MAX	How parallel envelope stages are combined: SUM or MAX.
COND	Trig condition	STEP: TRIG	ALL PROB A:B FILL !FIL 1ST !1ST PRE !PRE	When the step plays (Appendix C).
CR	Crop	WAVE: REC	four clicks	Trims the sample to its start and end markers.
CT	Category	PRESET: bar	list	Filters the list by category.
CURV	Shaping curve	Insert effect: WSH	SOFT HARD FOLD ASYM	SOFT, HARD, FOLD or ASYM transfer curve.
CURV	Envelope curve	Synth: F ENV, AMP; FM, Macro, Wavetable, Wave seq: F ENV	−100…+100	Shape of the envelope segments, from logarithmic through linear to exponential.
CUT	Cutoff (CC 74)	MIDI track: MIDI	0–127	Sends the filter cutoff controller to the external instrument.
CUT	Cutoff	Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT	varies	Filter cutoff frequency.
CUTB	Filter B cutoff	Synth: LFO DEST	destination	The cutoff of filter B alone, up to four octaves either way.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
DAMP	Damping	Insert effect: RVB, HALL, SPRG, SHIM	0–100	How quickly high frequencies die away in the tail.
DAMP	Damping (string models)	Macro: MACRO	0–100	Macro of the PLCK and BOW models: string damping.
DAMP	Damping	TRK FX: DIRT	0–100	Darkens the DIRT block's feedback.
DC	Decay	DRUMS: bar	0.01–4 s	DEC of the selected drum pad.
DCAY	Decay time	Insert effect: RVB, HALL, SPRG, SHIM	varies	Length of the reverb tail.
DEC	Decay	Drums: DR	0.01–4 s	Length of the drum voice. A drum rings for its DEC however short the pad press or the step; SUB sounds while held and fades over DEC afterwards.
DEC	Decay (CC 75)	MIDI track: SOUND	0–127	Sends the decay-time controller.
DEC	Decay	Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP	varies	Time the envelope takes to fall to the sustain level.
DECn	Envelope decay	Synth, FM: LFO DEST	destination	FM, DEC1 to DEC6: the decay of operator 1 to 6. Synth, DEC2 to DEC5: the decay of an extra envelope (MENV).
DENS	Density	ARP: VARY	0–100	How many steps sound.
DENS	Density (particle noise model)	Macro: MACRO	0–100	First macro of the DUST model: particle density.
DENS	Grain density	Sampler: GRAIN	1–200 Hz	Grains started per second.
DEST	Pressure destination	ARP: ACFG	RATE OCT GATE	What pad pressure does to the arp.
DEST	Route destination	AT ROUTE: top row	per engine	What pad pressure moves, for routes 1 to 4 of this track.
DEST	LFO destination	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2	varies	What the LFO moves; the list depends on the engine (Appendix C).
DET2	Oscillator 2 detune	Synth: OSC	−50…50 cents	Fine detune of oscillator 2 in cents, for beating and width.
DETn	Operator detune	FM: LFO DEST	destination	DET1 to DET6: the DETN cell of operator 1 to 6.
DETN	Operator detune	FM: OP1–OP6	−50…50 cents	Fine detune of this operator in cents.
DETN	Detune (FM model)	Macro: MACRO	0–100	Third macro of the FM model: detunes the modulator up to a semitone either way of 50; the tone beats and turns inharmonic.
DICE	Dice	ARP: VARY	0–100	Re-rolls the random choices behind the variations.
DIR	Strum direction	ARP: PLAY	UP DOWN ALT	Direction of the strum.
DIR	Scan direction	Wavetable: SCAN	FWD REV	FWD runs from POS to the end of the table; REV from the end back to POS.
DISP	Dispersion	Insert effect: SPRG	0–100	The spring's characteristic chirp.
DIV	LFO division	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2	17 values (App. C)	The tempo division used while SYNC is on: 8 bars to 1/32, with dotted and triplet values.
DL	Delete	PRESET, SONG: bar	four clicks	Deletes the user preset (PRESET) or the selected row (SONG).

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
DLY	LFO delay	Synth: MOD2	seconds	Wait after note-on before the LFO starts.
DPTH	Pressure depth	ARP: ACFG / ROLL	0–100	How far pressure moves it.
DPTH	Route depth	AT ROUTE: bottom row	−100…+100	How far; drums 0 to 100.
DPTH	Depth	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2; Insert effect: CHR, FLG, PHS, TRM, ENS, PAN, RING, VIB	varies	Amount of modulation. On LFOs negative values invert the shape.
DPTH	Vowel LFO depth	TRK FX: TALK	0–100	How far the vowel moves by itself.
DPTn	LFO depth	Synth: LFO DEST	destination	DPT1 to DPT4: the depth of LFO 1 to 4, from nothing to double.
DRAT	Drift rate	Synth: ANALG	0.05–5 Hz	Speed of the pitch drift.
DRFT	Drift amount	Synth: ANALG	0–100	Slow random wander of each oscillator's pitch.
DRIV	Drive	Drums: DR	0–100	Saturation of this drum voice, on every type. The snares also shift from body to noise with it, CLAP adds claps, SHKR rattles deeper.
DRIV	Drive	Sampler: TONE	0–100	Saturation of the sampler voice.
DRIV	Filter drive	Synth, FM, Macro, Wavetable, Wave seq: FILTR	0–100	Input drive into the filter. Each type overdrives in its own way; the SVF soft-clips its input, and on COMB it darkens the feedback.
DRIV	Mixer drive	Synth: OSC	0–100	Overdrives the oscillator mix before the filter.
DRIV	Drive	TRK FX: DIRT; Insert effect: DRV, WSH	0–100	Amount of distortion.
DRUM	Drum speed	Insert effect: ROT	0.5–2 Hz	Fast-speed rotation of the bass rotor.
DRVB	Filter B drive	Synth: FLT B	0–100	Input drive into filter B.
DST1	Strip route 1 destination	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	varies	STRIP page: what the mod strip moves on route 1. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group.
DST2	Strip route 2 destination	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	varies	STRIP page: what the mod strip moves on route 2. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group.
DST3	Strip route 3 destination	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	varies	STRIP page: what the mod strip moves on route 3. The same list as the LFOs' DEST (Appendix C); from the strip, LRAT and LDPT act on both LFOs. Func + turn goes from group to group.
DV	Drive	DRUMS: bar	0–100	DRIV of the selected drum pad.
ED	Edit	SCALE: bar	OFF / ON	Edit mode: tap a pad, then a key assigns its note.
EDPT	Extra envelope depth	Synth: LFO DEST	destination	The DPTH cell of the MENV page.
EN	Engine filter	PRESET: bar	THIS / ANY	Show presets for this track's engine only, or all.
EN	End	WAVE: WAVE	position	End marker (END).
END	End	Sampler: SAMPL	0–100	Where playback ends.
ENG	Engine	TRACKS: ENG	8 engines	Knob N chooses a pending engine for track N; that track's top pad commits it.
ENV	Envelope amount	Synth, FM, Macro, Wavetable, Wave seq: FILTR	−100…+100	How far the filter envelope moves the cutoff; negative closes the filter. Up to 7 octaves.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
ENV	Envelope to vowel	TRK FX: TALK	−100…+100	How far the playing level pushes the vowel; negative pushes it the other way.
ENVF	Envelope to filter	Sampler: FILTR	−100…+100	How far the amp envelope moves the cutoff; negative closes it.
ENVP	Envelope to position	Wavetable: WAVE	−100…+100	The amp envelope moves the table position.
ENVS	Envelope count	Synth: MENV	0–4	How many extra envelopes each voice runs.
EVAR	Envelope variation	Synth: ANALG	0–100	Random change of envelope times per note.
EVEL	Velocity to envelope amount	FM, Macro, Wavetable, Wave seq: F ENV	0–100	Harder notes increase the filter envelope's reach.
EVOD	Even / odd (harmonic model)	Macro: MACRO	0–100	Second macro of the HARM model: balance of even and odd harmonics.
EVOL	Evolution	ARP: MUTA	0–99	Generations mutated so far; anticlockwise returns to the ancestor (an undo step).
EXPR	Expression (CC 11)	MIDI track: MIDI	0–127	Sends the expression controller.
FADE	LFO fade-in	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2	0–5 s	Time the LFO takes to reach full depth after a note starts.
FADE	Fade-in	Insert effect: VIB	0–2 s	Time the vibrato takes to come in.
FADE	Slice fade	Sampler: SLICE	0–0.1 s	Fades the end of each slice to avoid clicks.
FATK	Filter envelope attack	Synth, FM, Macro, Wavetable, Wave seq: LFO DEST	destination	The ATK cell of the F ENV page.
FDBK	Operator feedback	FM: OP1–OP6	0–100	The operator modulates itself; adds bite, then noise.
FDBK	Feedback	Insert effect: FLG, PHS, DLY, RVSE, TAPE, LCR, MTAP	varies	How much of the effect's output is fed back into it.
FDBK	Feedback (plucked string and FM models)	Macro: MACRO	0–100	Third macro of the PLCK model: string feedback (sustain). Fourth macro of the FM model: the modulator feeds back into itself; bite, then noise.
FDBK	Feedback	TRK FX: DIRT	0–100	Tuned feedback around the distortion; howls at high settings.
FDBn	Operator feedback	FM: LFO DEST	destination	FDB1 to FDB6: the FDBK cell of operator 1 to 6.
FDEC	Filter envelope decay	Synth, FM, Macro, Wavetable, Wave seq: LFO DEST	destination	The DEC cell of the F ENV page.
FDRV	Filter drive	Sampler: FILTR	0–100	Input drive into the sampler's filter.
FFM	Filter FM	Synth: FILTR	0–100	Oscillator 2 modulates the cutoff at audio rate.
FILT	Smoothing filter	Insert effect: RED	0–100	Softens the reducer's aliasing.
FINE	Fine tune	Synth: OSC3; Macro: MACRO; Wavetable: OSC; Wave seq: SEQ; Sampler: TONE	varies	Fine pitch of the whole sound.
FL.D	Flanger depth	TRK FX: MODFX	0–100	Amount of flanger; 0 bypasses it.
FL.R	Flanger rate	TRK FX: MODFX	0.1–5 Hz	Speed of the block's flanger.
FMOR	Filter morph	Macro: LFO DEST	destination	The MORP cell of the FILTR page (the macro oscillator's own MORP is its fourth macro).
FOLD	Fold (wavefolder model)	Macro: MACRO	0–100	First macro of the FOLD model: fold amount.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
FREL	Filter envelope release	Synth, FM, Macro, Wavetable, Wave seq: LFO DEST	destination	The REL cell of the F ENV page.
FREQ	Carrier frequency	Insert effect: RING	20 Hz–5 kHz	Frequency the signal is multiplied with.
FREQ	Wah frequency	Insert effect: WAH	150 Hz–2.5 kHz	Centre of the wah's sweep.
FREZ	Freeze	ARP: PLAY	OFF / ON	Holds the current note order while you change keys.
FRMS	Frames	Wavetable: SCAN	64 128 256 A64 A128 A256	How a sample is cut into the USER table: 64, 128 or 256 equal pieces, or A64 / A128 / A256, aligned to the sound's pitch period so scanning is smooth and plays in tune.
FSUS	Filter envelope sustain	Synth, FM, Macro, Wavetable, Wave seq: LFO DEST	destination	The SUS cell of the F ENV page.
FVAR	Filter variation	Synth: ANALG	0–100	Random cutoff offset per note.
FVEL	Velocity to cutoff	Synth, FM, Macro, Wavetable, Wave seq: LFO DEST	destination	The VEL cell of the F ENV page.
GAIN	Gain	Insert effect: AMP	0–100	Amp preamp gain.
GAP	Minimum gap	Sampler: SLICE	0.01–1 s	Shortest time allowed between two detected hits; raise it to merge flams and rolls into one slice.
GATE	Gate	ARP: ARP / ROLL	0–100	Length of each arp note (or roll repeat) as a share of its step.
GATE	Arp gate lock	TRK FX: ARPLK	0–100	Per-step lock of the arpeggiator's gate length.
GLID	Glide	Synth: VOICE; Wavetable: OSC	0–2 s	Portamento time between notes.
GLMD	Glide mode	Synth: VOICE	ALL LEGA	ALL: every note glides. LEGA: only overlapping notes glide.
GPOS	Grain position	Sampler: LFO DEST	destination	GRAN mode: where new grains start, around the scan position.
GVAR	Gate variation	ARP: VARY	0–100	Spread of gate lengths.
HARM	Harmonics (virtual analog model)	Macro: MACRO	0–100	Third macro of the VA model. 50 is the saws as they are; below, each fades into a sine; above, a saw an octave up is mixed in.
HARM	Highest harmonic	Wavetable: KEYS	1–127	The highest harmonic kept of every key of the KEYS table, 1 to 127.
HCUT	High cut	Insert effect: RVSE	500 Hz–20 kHz	Darkens the reversed echoes.
HELP	Help cards	SETTINGS: SETUP	OFF 0.35S INST	When holding Shift or Func alone shows its help card: never, after 0.35 seconds (the default), or at once. Kept between sessions.
HIGH	High gain	Insert effect: EQ, GEQ	varies	Boost or cut of the high band.
HIGH	Master EQ high	SETTINGS: MASTR	±18 dB	High band of the master EQ.
HMID	High-mid gain	Insert effect: GEQ	−12…12 dB	Boost or cut of the high-mid band.
HOLD	Kick hold	Drums: KICK	0–0.3 s	Kick machine: time at full level before the decay starts.
HORN	Horn speed	Insert effect: ROT	5–10 Hz	Fast-speed rotation of the treble horn.
HUMN	Humanize	ARP: PLAY / ROL2	0–100	Random variation of timing and velocity.
IN	Insert row	SONG: bar	turn	Inserts a row.
INDX	Index (FM model)	Macro: MACRO	0–100	First macro of the FM model: modulation depth.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
INTP	Interpolation	Wavetable: WAVE	LIN SPEC STEP VINT CLSC	How the table is read: LIN smooth, SPEC spectral, STEP stepped, VINT vintage (gritty), CLSC classic: as an early digital wavetable synth read its tables (64 positions, waves of 128 samples and 8 bits, nothing smoothed or band-limited), which gives high notes a glassy, aliased top.
INTV	Mutate interval	ARP: MUTA	1–16	Bars between mutations.
JIT	Jitter	Sampler: STRCH	0–100	Randomises where each cycle or grain starts; smooths the buzz of CYCL, widens the blur of SMER.
KEY	Key follow	Sampler: TONE	OFF ON	On: pitch follows the note. Off: every note plays the sample at its original pitch.
KEY	Key tracking	Synth: F ENV	0–100	How far the cutoff follows the played note.
KFRM	Key table frames	Wavetable: KEYS	8 16 32 64 128 256	How many frames the KEYS table has: 8 to 256. Few frames and INTP on STEP or CLSC make the steps of a scan audible.
KM	Key mapping	SCALE: bar	root / number	Keys pick a page by its root or by its number.
KNCK	Knock	Drums: KICK	0–100	Kick machine: level of the short tick at the very start of the drum.
KS	Keys select pages	SCALE: bar	OFF / latched / momentary	Keyboard keys choose the scale page.
KTON	Knock tone	Drums: KICK	0–100	Kick machine: from a dull thud of about 12 ms to a bright tick of about 1.5 ms.
KTRK	Key tracking	FM, Macro, Wavetable, Wave seq, Sampler: FILTR	0–100	How far the cutoff follows the played note.
KY	Key frame	WAVE: WAVE (wavetable track)	1–8	Which key frame of the KEYS table the page shows and edits. The top-row pads choose it too.
L1–L16	Step level	Wave seq: LVL A / B	0–100	LVL A / LVL B pages: level of steps 1 to 16.
>LCK	Pressure to locks	STEP: PRES	four clicks	Turns the step's recorded pressure into parameter locks on the steps the note spans, through the track's pressure routes as they are now, then removes the recording.
LCKS	Locks	STEP: TRIG	count	Number of parameter locks on the step; turning it anticlockwise clears them (an undo step).
LD	Load	PRESET: bar	four clicks	Keeps the highlighted preset as the track's sound for good.
LDPT	Other LFO's depth	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO DEST	destination	The depth of the other LFO, from nothing to double.
LE	Loop end	WAVE: WAVE	position	Loop end marker (LPEN).
LEGA	Legato	Synth: VOICE	OFF ON	On: overlapping notes do not retrigger the envelopes.
LEN	Note length	STEP: TRIG	0.25–16 steps	How long the step's note is held.
LEN	Stretch length	Sampler: STRCH	10 values (App. C)	With SYNC on, how long the loop lasts: 1/16 of a bar to 32 bars.
LEN	Track length	TRK FX, STEP: TRACK / bar	1–1024	Number of steps in this track's pattern; every track can differ (polymeter). STEP shows it as LN.
LEN	Sequence length	Wave seq: SEQ	1–16	Number of steps in the wave sequence.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
LEN	Scan length	Wavetable: SCAN	1/16 1/8 1/4 1/2 1BAR 2BAR 4BAR 8BAR 16BR	With SYNC on, how long one pass lasts: 1/16 to 16 bars.
LEVL	Level	Drums: DR	0–100	Level of this drum voice.
LEVL	Operator level	FM: OP1–OP6	0–100	Output level of this operator: loudness for a carrier, modulation depth for a modulator.
LEVL	Output level	Insert effect: AMP	−24…12 dB	Output level of the amp effect in dB.
LEVL	Level	Synth: AMP; FM: FMMIX; Macro: MACRO; Wavetable: WAVE; Wave seq: SEQ; Sampler: TONE	0–100	Output level of the engine, before the track's mixer volume.
LFO	LFO select	Synth: MOD / MOD2	LFO1–LFO4	Which LFO the page edits.
LFOS	LFO count	Synth: MOD	1–4	How many LFOs each voice runs.
LK	Scale lock	SCALE: bar	OFF / ON	Forces keys and arps into the scale (also the Scale button).
LLEN	Level lane length	Wave seq: LANES	1–16	Length of the level lane. Lanes of different lengths drift against each other.
LMID	Low-mid gain	Insert effect: GEQ	−12…12 dB	Boost or cut of the low-mid band.
LN	Track length	STEP: bar	1–1024	Length of the selected track in steps.
LN	Length	WAVE: REC	free, 1 beat … 8 bars	Length of the take.
LOAD	Load project	SETTINGS: PROJ	four clicks	Loads the slot. Shows OK or FAIL.
LOCK	Macro / lock interaction	ARP: ACFG	ADD MACRO LOCK MULT	How variation macros and step locks combine.
LOOK	Look-ahead	Insert effect: LMT	1–5 ms	How far ahead the limiter looks, in milliseconds.
LOOP	Loop mode	Wave seq: SEQ	FWD REV PING RAND ONCE	FWD, REV, PING, RAND or ONCE.
LOW	Low gain	Insert effect: EQ, GEQ	varies	Boost or cut of the low band.
LOW	Master EQ low	SETTINGS: MASTR	±18 dB	Low band of the master EQ.
LP	Loop	SONG: bar	OFF / ON	Loops the song at the end.
LPEN	Loop end	Sampler: SAMPL	0–100	End of the looped region.
LPST	Loop start	Sampler: SAMPL	0–100	Start of the looped region.
LRAT	Other LFO's rate	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO DEST	destination	The rate of the other LFO, up to two octaves either way.
LS	Loop start	WAVE: WAVE	position	Loop start marker (LPST).
LTCH	Latch	ARP: ARP / ROL2	OFF / ON	ARP: the arp keeps playing after the keys are released (also Shift + Arp). ROL2: a pad press starts a roll that runs until pressed again.
LV	Level	DRUMS: bar	0–100	LEVL of the selected drum pad.
LVL1–LVL6	Operator level	FM: FMMIX	0–100	FMMIX page: output level of operators 1 to 6, one knob each.
LVL3	Oscillator 3 level	Synth: OSC3	0–100	Level of oscillator 3 (sub).
LVLn	Oscillator level	Synth: LFO DEST	destination	LVL1 to LVL3: the level of one oscillator, from silence to double.
MAKE	Make-up gain	Insert effect: CMP	0–30 dB	Gain after compression.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
MAKE	Master make-up gain	SETTINGS: MASTR	0–30 dB	Gain after the master compressor.
MAX	Maximum rate	ARP: ROLL	division or Hz	Repeat rate at full pressure.
MD	Mode	SCALE: bar	RL / AB	Relative (pads follow the root) or absolute.
MD	Play mode	WAVE: WAVE	FWD … GRAN	The sampler's MODE.
MDPT	Motion depth	Wavetable: WAVE	0–100	Depth of the built-in position sweep.
MFRQ	Mid frequency	Insert effect: EQ	100 Hz–8 kHz	Centre frequency of the mid band.
MFRQ	Master EQ mid frequency	SETTINGS: MASTR	100 Hz–8 kHz	Centre of the mid band.
MICR	Micro-timing	STEP: TRIG	−50…+50 %	Moves the step earlier or later by up to half a step.
MID	Mid gain	Insert effect: EQ	−18…18 dB	Boost or cut of the mid band.
MID	Master EQ mid	SETTINGS: MASTR	±18 dB	Mid band of the master EQ.
MIX	Mix (particle noise model)	Macro: MACRO	0–100	Fourth macro of the DUST model: balance of the noise particles against the tone.
MIX	Oscillator mix	Synth: OSC	0–100	Balance between oscillator 1 and oscillator 2.
MIX	Dry / wet mix	TRK FX: DIRT, TALK; Insert effect: CHR, DRV, FLG, DLY, RVB, ENS, RED, RVSE, RING, TAPE, LCR, MTAP	0–100	Balance between the untreated sound (0) and the effect (100).
MOD	Modulation (CC 1)	MIDI track: MIDI	0–127	Sends the mod-wheel controller.
MODB	Filter B mode	Synth: FLT B	---	Response of the second filter.
MODE	Arp mode	ARP: ARP	12 modes	Order the held notes play in (Appendix C).
MODE	Delay mode	Insert effect: DLY	STER PING TAPE	STER stereo, PING ping-pong, TAPE with tape-style tone.
MODE	Shimmer interval	Insert effect: SHIM	OCT 5TH O+5	Pitch shift fed into the reverb: octave, fifth, or both.
MODE	Play mode	Sampler: SAMPL	FWD REV 1SHT GATE LOOP PING GRAN	FWD forward, REV reverse, 1SHT one-shot, GATE while held, LOOP, PING back-and-forth loop, GRAN granular.
MODE	Slice mode	Sampler: SLICE	GRID TRNS	GRID: equal slices. TRNS: one slice per detected hit (transient).
MODE	Filter mode	Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT	LP BP HP NTCH	Response of the chosen filter type (low-pass, band-pass, high-pass, notch, slopes); the list depends on TYPE.
MODE	Voice mode	Synth: VOICE	MONO DUO POLY UNI	MONO, DUO (two voices), POLY, or UNI (all voices stacked on one note).
MODE	Scan mode	Wavetable: SCAN	OFF RUN LOOP PING STEP RAND	Moves the table position by itself over a set time: OFF, RUN (once, then holds), LOOP, PING (there and back), STEP (in audible steps), RAND (random frames).
MODL	Model	Macro: MACRO	VA FM FOLD FORM HARM PLCK DUST BOW	Synthesis model: VA, FM, FOLD, FORM, HARM, PLCK, DUST, BOW. Knobs 2 to 5 are renamed to suit the model.
MORP	Morph	Macro: MACRO, FILTR	0–100	Fourth macro of the VA, FORM and PLCK models. VA: saw to square. FORM: voice to whisper (noise takes over from the pulse train). PLCK: string to drum (the pitch dissolves into a burst).

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
MORP	Filter morph	Synth, FM, Wavetable, Wave seq: FILTR	0–100	On the SEM type in MRPH mode: sweeps low-pass through notch to high-pass. Anywhere else the knob says SEM MRPH ONLY.
MSPD	Motion speed	Wavetable: WAVE	0–10 Hz	Speed of the built-in sweep of the table position.
MUTA	Mutate	ARP: MUTA	OFF / ON	Switches mutation on: the arp pattern changes by itself over time. SPCE on the same page sets how much.
MVOL	Master volume	SETTINGS: SETUP	0–100	Overall output level.
nA	Oscillator source, first cell	Synth: LFO DEST	destination	1A, 2A, 3A: the first SRC cell of oscillator 1, 2 or 3 (table position, FM index, formant frequency or harmonic tilt, by what the slot plays).
nB	Oscillator source, second cell	Synth: LFO DEST	destination	1B, 2B, 3B: the second SRC cell of oscillator 1, 2 or 3 (table, FM ratio, formant width or odd / even balance); tables and ratios change in steps.
NCOL	Noise colour	Synth: OSC3	0–100	Tone of the noise, from dark to bright.
NEW	New project	SETTINGS: PROJ	four clicks	Starts an empty project.
nFRM	Formant frequency	PARAM: SRC	0–100	Synth, slot n set to FORM: the resonant peak, from 1 to 16 times the note's pitch.
nIDX	FM index	PARAM: SRC	0–100	Synth, slot n set to FM: how hard the modulator pushes the carrier.
NM	Normalise	WAVE: REC	OFF / ON	Raises the take to full level.
nODD	Odd harmonics	PARAM: SRC	0–100	Synth, slot n set to HARM: from all harmonics to odd ones only.
NOIS	Noise level	Synth: OSC3	0–100	Level of the noise source.
NORM	Normalise	Sampler: SAMPL	OFF ON	On: the sample's loudest point is brought to full level.
NORM	Frame level	Wavetable: KEYS	OFF PEAK RMS	Level of the frames of the KEYS table. OFF: as computed. PEAK: every frame at full scale. RMS: every frame equally loud.
NOTE	Step note	STEP: TRIG	C-1 … G9	Pitch of the step; a chord moves together.
nPOS	Wavetable position	PARAM: SRC	0–100	Synth, slot n set to WT: where in the table the oscillator reads.
nRAT	FM ratio	PARAM: SRC	1–12	Synth, slot n set to FM: the modulator's pitch as a whole multiple of the note.
NT	Note	SCALE: bar (edit)	note	Edit mode: the selected pad's note.
nTBL	Wavetable table	PARAM: SRC	SAW … DIGI, VOWL … COMB	Synth, slot n set to WT: which of the wavetable engine's 23 built-in tables (the first nine, then the fourteen newer ones).
nTLT	Harmonic tilt	PARAM: SRC	0–100	Synth, slot n set to HARM: from all harmonics equally loud (bright) to steeply falling (dark).
NUM	Number (harmonic model)	Macro: MACRO	0–100	Third macro of the HARM model: number of harmonics.
nWID	Formant width	PARAM: SRC	0–100	Synth, slot n set to FORM: how broad the peak is.
OC	Octave	SCALE: bar	octaves	Octave of the pads.
OCT	Octave range	ARP: ARP	1–8	Octaves the arp spans.
OCT	Octave	Wavetable: OSC	−4…4	Octave shift of the oscillator.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
OFF	No destination	all engines: LFO DEST	destination	The LFO does nothing and costs nothing.
OFSB	Filter B offset	Synth: FLT B	−48…48 semitones	Cutoff of filter B relative to filter A, in semitones.
ON	Insert on	TRK FX: FX	OFF / ON	Switches the slot on or off. Effects with a TIME or RATE show SYNC here instead.
OUT	Output gain	Insert effect: TRNS	−12…12 dB	Output level in dB.
P1–P16	Step pitch	Wave seq: PIT A / B	−24…24 semitones	PIT A / PIT B pages: pitch offset of steps 1 to 16, one knob per step.
PA	Pattern	SONG: bar	1–16	Pattern this row plays.
PAN	Pan (CC 10)	MIDI track: MIDI	0–127	Sends the pan controller.
PAN	Pan	Sampler: AMP; Drums: DR	varies	Left / right position.
PAN	Track pan	TRACKS: PAN	L–R	Mixer page: knob N is the pan of track N.
>PAT	Write to pattern	ARP: MUTA	four clicks	Writes the arp's current steps into the track's pattern (an undo step).
PAT	Pattern	SETTINGS: SETUP	1–16	Selects the pattern (same as > + pad).
PD	Pad	SCALE: bar (edit)	1–16	Edit mode: which pad is being edited.
PDLY	Pre-delay	Insert effect: HALL	0–100 ms	Gap before the reverb starts.
PG	Scale page	SCALE: bar	12 factory + user	Which pad layout (scale) the pads play.
PHAS	Stereo phase	Insert effect: PAN	0–100	Phase offset between left and right of the auto-pan.
PHAS	Random phase	Synth: ANALG	OFF ON	On: oscillators start at a random point in their cycle on each note.
PH.D	Phaser depth	TRK FX: MODFX	0–100	Amount of phaser; 0 bypasses it.
PH.R	Phaser rate	TRK FX: MODFX	0.1–5 Hz	Speed of the block's phaser.
PICK	Pick (plucked string model)	Macro: MACRO	0–100	Second macro of the PLCK model: pick position.
PITn	Oscillator 2 pitch	Synth: LFO DEST	destination	PIT2: the pitch of oscillator 2 alone, up to an octave either way.
PJIT	Pitch jitter	Sampler: GRAIN	0–100	Random pitch per grain, up to an octave either way.
PL	Key place	WAVE: WAVE (wavetable track)	0–100	Where along the table the key sits (PLCn).
PLAY	Slice play	Sampler: SLICE	ONE KEYS NEXT RAND	ONE: the slice chosen by SLIC. KEYS: each key plays its own slice, upward from ROOT. NEXT: each new note plays the next slice. RAND: a random slice per note.
PLC1–PLC8	Key place	Wavetable: K PLC	0–100	K PLC page: where along the table key frame 1 to 8 sits. Only order and spacing count: the earliest key in use is the first frame, the latest the last.
PLEN	Pitch lane length	Wave seq: LANES	1–16	Length of the pitch lane.
PMOD	Poly-mod	Synth: F ENV	−100…+100	Filter envelope to oscillator 2 pitch, up to two octaves either way.
PN	Pan	DRUMS: bar	L–R	PAN of the selected drum pad.
PO	Key position	WAVE: WAVE (wavetable track)	position	Where in the sample the key's cycle is taken (POSn): the line on the waveform.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
POS	Position (bowed string model)	Macro: MACRO	0–100	Second macro of the BOW model: bow position.
POS	Table position	Wavetable: WAVE	0–100	Where in the wavetable the oscillator reads; sweeping it morphs the tone.
POS1–POS8	Key position	Wavetable: K POS	0–100	K POS page: where in its sample key frame 1 to 8 is taken. One cycle is cut there (one period where the sample has a pitch) and plays at the pitch of the keys.
PRES	Pressure (bowed string model)	Macro: MACRO	0–100	First macro of the BOW model: bow pressure, from a light, glassy bow to a pressed, rough one.
PRES	Recorded pressure amount	STEP: PRES	0–200 %	How much of the pad pressure recorded with the step's notes is played back. 0 plays none and keeps the recording; hold > and turn for 100.
PRNG	Pitch range	Wave seq: SEQ	0–2×	Scales how far the pitch lane reaches (0 = flat, 2 = double).
PROB	Repeat probability	ARP: ROL2	0–100	Chance each repeat sounds.
PROB	Probability	STEP: TRIG	0–100 %	Chance the step plays, with COND = PROB.
PROG	Program	MIDI track: MIDI	0–127	Sends a program change.
PS	Preset	PRESET: bar	list	Scrolls the preset list.
PTCH	Pitch	Sampler: TONE	−24…24 semitones	Transposition of the sample in semitones.
PTCH	Pitch	all engines: LFO DEST	destination	Pitch of the whole voice, up to an octave either way; the lower half of DPTH is vibrato territory.
P>V	Pressure to velocity	ARP: ROLL	0–100	How much pressure sets each repeat's velocity.
PW	Pulse width / shape	Synth: OSC	5–95	Shapes both oscillators: 50 is the plain wave, and the two directions are two different sounds. Below 50 thins or sharpens: SQR and PLS narrow to a thin pulse, SIN and TRI are wavefolded, SAW gains a second, shifted saw. Above 50 thickens: SQR and PLS gain a square an octave up, SIN and TRI are pushed towards a square, SAW hollows into a square.
Q	Q (formant model)	Macro: MACRO	0–100	Third macro of the FORM model: formant sharpness.
QN	Playback quantize	STEP: bar	0–100	Pulls recorded notes toward the grid on playback; the recording itself is never changed.
QN	Start quantize	WAVE: REC	NOW / BAR	Start at once, or on the next bar.
QNT	Quantize	SETTINGS: SETUP	0–100	Global playback quantize.
RAND	Random start	Sampler: AMP	0–100	Each note starts at a random point up to this far into the sample.
RAT1–RAT6	Operator ratio	FM: RATIO	0.5–16×	RATIO page: frequency ratio of operators 1 to 6, one knob each.
RATE	Arp rate	ARP: ARP	1/1–1/64T	Note value of the arp's steps.
RATE	Rate	FM, Macro, Wavetable, Sampler, Drums: LFO1 / LFO2; Insert effect: CHR, FLG, PHS, TRM, ENS, PAN, RING, VIB, WAH	varies	Speed of the LFO or effect modulation. With SYNC on it picks a tempo division instead.
RATE	Sample-rate reduction	Sampler: TONE	0–100	Lowers the playback sample rate for aliasing grit.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
RATE	Sample-rate reduction	TRK FX: DIRT; Insert effect: RED	0–100	Lowers the sample rate for aliasing grit.
RATE	Vowel LFO rate	TRK FX: TALK	0.05–12 Hz	Speed of the vowel movement.
RATE	Rate	Wave seq: SEQ, LFO1 / LFO2	varies	Speed of the wave sequence relative to the tempo.
RATI	Operator ratio	FM: OP1–OP6	0.5–16×	Frequency of this operator as a multiple of the note.
RATI	Compression ratio	Insert effect: CMP, DUCK	1–20×	How strongly level above the threshold is reduced.
RATI	Ratio (FM model)	Macro: MACRO	0–100	Second macro of the FM model: modulator ratio.
RATI	Master compressor ratio	SETTINGS: MASTR	1–20	Strength of the master compression.
RATn	LFO rate	Synth: LFO DEST	destination	RAT1 to RAT4: the rate of LFO 1 to 4, up to two octaves either way.
RC	Record	WAVE: REC	arm / stop	Two clicks clockwise within a second arm; one anticlockwise stops.
RCHN	Ratchet chance	ARP: VARY	0–100	Chance of a step ratcheting.
RDEC	Ratchet decay	ARP: PLAY	0–100	How much each ratchet repeat is quieter.
REC	Record mode	ARP: ACFG	NOTE / CHORD	How arp output is written when recording.
REL	Release	Insert effect: LMT	50–500 ms	How fast the limiter recovers, in milliseconds.
REL	Release (CC 72)	MIDI track: SOUND	0–127	Sends the release-time controller.
REL	Release	Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP	varies	Time the envelope takes to fall to silence after the note ends.
RELn	Envelope release	Synth, FM: LFO DEST	destination	FM, REL1 to REL6: the release of operator 1 to 6. Synth, REL2 to REL5: the release of an extra envelope (MENV).
RELS	Release	Insert effect: DUCK	10–1000 ms	How fast the track comes back after ducking.
RESB	Filter B resonance	Synth: FLT B	0–100	Resonance of the second filter.
RESO	Resonance (CC 71)	MIDI track: MIDI	0–127	Sends the resonance controller to the external instrument.
RESO	Resonance	Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT, WAH	0–100	Emphasis at the cutoff frequency. The LADR type self-oscillates at full resonance.
RESO	Formant resonance	TRK FX: TALK	0–100	Sharpness of the vowel formants.
RHYT	Rhythm	ARP: ACFG	OFF / ON	Plays the arp in the tapped rhythm.
RING	Ring (particle noise model)	Macro: MACRO	0–100	Third macro of the DUST model, neutral at 50. Above: the particles ring at the pitch of the key. Below: a high-pass thins them into a hiss that follows the key.
RO.D	Rotary depth	TRK FX: MODFX	0–100	Amount of rotary speaker; 0 bypasses it.
ROLL	Roll	ARP: ROLL	OFF / ON	Hold-to-ratchet for every pad that plays a note.
ROOT	Root note	Sampler: SAMPL	0–127 note	The key at which the sample plays at its original pitch; in KEYS slice play, the key that plays slice 1.
RO.S	Rotary speed	TRK FX: MODFX	0–100	Speed of the rotary speaker.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
ROUT	Filter routing	Synth: FLT B	SNGL SER PAR	SNGL: filter A only. SER: A into B. PAR: A and B side by side.
RP	Repeats	SONG: bar	count	Times the row repeats.
RPT	Repeats	ARP: ARP	1–8	Times each note is repeated before moving on.
RPTS	Repeat limit	ARP: ROL2	0–16	Stops after this many repeats; 0 = no limit.
RT	Root	SCALE: bar	C–B	Root note of the scale.
RTCH	Ratchet	ARP: PLAY	1–4	Repeats inside each arp step.
RTCH	Arp ratchet lock	TRK FX: ARPLK	1–4	Per-step lock of the arpeggiator's ratchet count.
RVB	Reverb send (CC 91)	MIDI track: SOUND	0–127	Sends the reverb-level controller.
RVRS	Reverse chance	Sampler: GRAIN	0–100	Proportion of grains played backwards.
RW	Row	SONG: bar	row	Selects the row (with fewer than two rows there is nothing to select, and the name bar says so).
S1–S16	Step shape	Wave seq: SHP A / B	SIN SAW SQR TRI PLS NOIS	SHP A / SHP B pages: waveform of steps 1 to 16.
SAVE	Save project	SETTINGS: PROJ	four clicks	Saves to the current slot.
SAWS	Saws (virtual analog model)	Macro: MACRO	0–100	First macro of the VA model: number and blend of stacked saws.
SC	Scroll	WAVE: WAVE	position	Scrolls the zoomed view (zoom in with ZM first).
SCAN	Scan speed	Sampler: GRAIN	−2…2×	How fast the grain read position moves through the sample, independent of pitch. 0 freezes; negative runs backwards.
SCAT	Octave scatter	ARP: VARY	0–100	Chance of notes jumping octave.
SCRV	Kick sweep curve	Drums: KICK	−100…+100	Kick machine: where in the sweep time the fall happens; negative = nearly all at the start.
SENS	Sensitivity	Insert effect: WAH	0–100	How far the playing level opens the wah (auto-wah).
SENS	Sensitivity	Sampler: SLICE	0–100	How readily a hit is detected in TRNS mode; higher finds quieter hits.
SG	Song on	SONG: bar	OFF / ON	Arms song mode: Play then follows the rows.
SHAP	Kick shape	Drums: KICK	0–100	Kick machine: the body's waveform, from a sine through a triangle to a squared-off wave.
SHAP	Grain shape	Sampler: GRAIN	HANN TRI PERC SWEL	Grain window: HANN smooth, TRI, PERC percussive, SWEL swelling.
SHFT	Shift (formant model)	Macro: MACRO	0–100	Second macro of the FORM model: formant shift.
SHFT	Formant shift	TRK FX: TALK	−100…+100	Moves all formants down or up: larger or smaller mouth.
SHIM	Shimmer amount	Insert effect: SHIM	0–100	How much pitch-shifted signal is fed into the tail.
SHP∗	Shape, all steps	Wave seq: LANES	SIN SAW SQR TRI PLS NOIS	Sets every step's wave at once.
SHPE	LFO shape	FM, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2	SIN TRI SQR SAW RND SRND RAMP EXPU EXPD	LFO waveform: SIN TRI SQR SAW RND (stepped random) SRND (smooth random) RAMP EXPU EXPD.
SHPE	LFO shape	Insert effect: TRM, PAN	SIN TRI SQR	Waveform of the effect's own LFO.
SHPE	Shape (wavefolder model)	Macro: LFO1 / LFO2, MACRO	varies	Second macro of the FOLD model: input waveform.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
SIZE	Room size	Insert effect: RVB, HALL	0.25–2×	Size of the reverb space.
SIZE	Grain size	Sampler: GRAIN	0.005–0.5 s	Length of each grain.
SIZE	Cycle / grain size	Sampler: STRCH	0.005–0.5 s	Length of the pieces the stretch works in. Short is buzzy and tight, long is smooth and smeared.
SL	Slices	WAVE: WAVE	1–64	Number of slices (SLCS).
SLCS	Slices	Sampler: SLICE	1–64	GRID mode: number of equal slices. TRNS mode: shows how many hits were found; turning it down keeps only the strongest (1 = no limit, up to 64).
SLEN	Slice length	Sampler: SLICE	0–100	Shortens every slice to this share of its length, for gated, choppy playback.
SLEN	Shape lane length	Wave seq: LANES	1–16	Length of the wave-shape lane.
SLIC	Slice select	Sampler: SLICE	0–100	Which slice plays.
SLID	Slide	Wave seq: LANES	OFF ON	On: lane values glide between steps instead of jumping.
SLOP	Tuning slop	Synth: ANALG	0–50 cents	Each new note starts slightly out of tune by up to this many cents.
SLOT	Project slot	SETTINGS: PROJ	P01–P32	Which project slot LOAD, SAVE and SVAS act on.
SLOT	Insert slot	TRK FX: FX	slot number	Which of the track's insert slots the page shows.
SM	Sample select	WAVE: REC	list	Chooses the track's sample, or the USER wavetable's source.
SM	Key sample	WAVE: WAVE (wavetable track)	OFF, list	The sample this key is taken from (SMPn). Giving a key a sample selects the KEYS table.
SMP1–SMP8	Key sample	Wavetable: K SMP	19 values (App. C)	K SMP page: the sample key frame 1 to 8 is taken from. OFF: the key is not used.
SMPL	Sample	Sampler: SAMPL	18 values (App. C)	Which sample the track plays: factory hits and loop, your files, the drum breaks (B080 … B170, named by tempo), and resampled takes. Choosing a break sets LEN to its length in bars.
SN	Snap	DRUMS: bar	0–100	SNAP of the selected drum pad.
SNAP	Snap	Drums: DR	0–100	The front edge of the hit. 50 leaves the drum as it is; below, the onset is softened; above, the first milliseconds are lifted and the tonal drums get a click. TOM and KSYN set their attack time with it, CLAP also the length of each clap.
SND1	Send 1	TRACKS, TRK FX: SND1 / TRACK	0–100	Level sent to bus 1 (reverb by default).
SND2	Send 2	TRACKS, TRK FX: SND2 / TRACK	0–100	Level sent to bus 2 (delay by default).
SOFT	Soft start	Synth: ANALG	0–50 cents	Pitch overshoot at the start of each note, in cents.
SP	Track speed	STEP: bar	1/4 … 4x	Speed of the selected track.
SPCE	Spice	ARP: VARY / MUTA	0–100	Variation macro: how adventurous the variations are.
SPD	Speed	Insert effect: ROT	0–100	Rotary speaker speed, slow to fast.
SPD	Detector speed	Insert effect: TRNS	0–100	How quickly the transient shaper reacts.
SPD	Speed (bowed string model)	Macro: MACRO	0–100	Fourth macro of the BOW model: bow speed.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
SPD	Stretch speed	Sampler: STRCH	0.25–4×	Playback speed with the pitch kept (except TAPE): 0.25× to 4×. In SMER the speed is a sixteenth of the value shown.
SPD	Track speed	TRK FX, STEP: TRACK / bar	1/4 1/2 1x 2x 4x	Playback speed of the track against the tempo. STEP shows it as SP.
SPD	Scan speed	Wavetable: SCAN	0.25–4×	Divides the scan's pass time: 2× runs through twice as fast.
SPRD	Stereo spread	Insert effect: VIB, LCR	0–100	How far the effect is spread across left and right.
SPRD	Spread (virtual analog model)	Macro: MACRO	0–100	Second macro of the VA model: detune between the stacked saws.
SPRD	Grain spread	Sampler: GRAIN	0–100	Stereo spread of the grains.
SPRD	Stereo spread	Sampler: STRCH	0–100	Spreads the stretch's grains across left and right (GRAN and SMER).
SPRD	Unison spread	Synth: VOICE	0–100	Detune between the stacked unison voices.
SPRD	Spread	Wavetable: OSC	0–100	Detune and width of the wavetable's doubled voices.
SPRY	Spray	Sampler: GRAIN	0–100	Random scatter of grain positions and timing.
SR	Source	WAVE: REC	MSTR, T1 …	What is recorded: the master output or one track after its effects.
SRC	Sidechain source	Insert effect: DUCK	17 values (App. C)	Which signal makes the track duck: SELF or another track T1 to T16.
SSIZ	Stretch size	Sampler: LFO DEST	destination	The SIZE cell of the STRCH page (SIZE alone is the grain size).
SSPR	Stretch spread	Sampler: LFO DEST	destination	The SPRD cell of the STRCH page (SPRD alone is the grain spread).
ST	Start	WAVE: WAVE	position	Start marker (the sampler's STRT).
STCK	Unison stack	Synth: VOICE	OFF 2 4 8	Number of detuned copies stacked on every note (off, 2, 4, 8).
STEP	Scan steps	Wavetable: SCAN	4 8 16 32 64	Number of plateaus (STEP mode) or of new random frames per pass (RAND mode).
STER	Stereo	Insert effect: TRM	0–100	Offsets left and right so the tremolo moves across the stereo field.
STGS	Stages	Insert effect: PHS	2–6	Number of phaser stages.
STGS	Chain stages	Macro, Wavetable, Wave seq: AMP	varies	Number of envelopes in the chain.
STRC	Stretch (harmonic model)	Macro: MACRO	0–100	Fourth macro of the HARM model: spreads the harmonics apart, as on a stiff string or a bar; the tone turns into a bell.
STRM	Strum	ARP: PLAY	OFF / ON	Plays held notes as a strummed chord.
STRP	Strip target	ARP: VARY	OFF SPCE DICE DENS SCAT RCHN GVAR GATE	Which macro the mod strip drives while this track's arp is on.
STRT	Start	Sampler: SAMPL	0–100	Where playback starts in the sample.
SUB	Sub level	Wavetable: OSC	0–100	Level of the sub oscillator one octave down.
SUS	Sustain gain	Insert effect: TRNS	−100…+100	Boosts or reduces the tail after each hit.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
SUS	Sustain	Synth: F ENV, AMP; FM: OP1–OP6, F ENV; Macro, Wavetable, Wave seq: AMP, F ENV; Sampler: AMP	0–100	Level held while the note is down.
SUSn	Envelope sustain	Synth, FM: LFO DEST	destination	FM, SUS1 to SUS6: the sustain of operator 1 to 6. Synth, SUS2 to SUS5: the sustain of an extra envelope (MENV).
SV	Save	PRESET: bar	four clicks	Saves the track's sound as a user preset (User 1, User 2, ...); the Mac then asks for a name.
SVAS	Save as	SETTINGS: PROJ	four clicks	Saves to the slot chosen with SLOT.
SWEP	Kick sweep	Drums: KICK	0–72 semitones	Kick machine: how far above TUNE the pitch starts, in semitones.
SWNG	Swing	SETTINGS, ARP: SETUP / ARP / ROL2	0–100	Delays every other step. SETUP: the pattern's swing. ARP and ROL2: the arp's and the roll's own.
SWPT	Kick sweep time	Drums: KICK	0.002–1 s	Kick machine: how long the pitch takes to fall to TUNE.
SYNC	LFO tempo sync	FM, Macro, Wavetable, Sampler, Drums: LFO1 / LFO2	OFF ON	On: RATE (or DIV) is a tempo division instead of Hz.
SYNC	Stretch tempo sync	Sampler: STRCH	OFF ON	On: the loop is stretched to last exactly LEN at the current tempo, and follows tempo changes.
SYNC	Tempo sync	Wave seq: LANES, LFO1 / LFO2	OFF ON	On: step times follow the tempo.
SYNC	Scan sync	Wavetable: SCAN	OFF ON NAT	OFF: the pass lasts TIME. ON: it lasts LEN at the tempo. NAT: it lasts as long as the original recording (USER table).
T1–T16	Step time	Wave seq: TIM A / B	0.03–2 s	TIM A / TIM B pages: duration of steps 1 to 16.
TABL	Table	Wavetable: WAVE	25 values (App. C)	Which wavetable: 23 built-in tables, USER (a sample cut into frames) or KEYS (a few waves picked out of samples, the frames between them computed).
TAP	Tap rhythm	ARP: ACFG	OFF / ON	While on, every pad press taps a step of a custom rhythm.
TAPS	Taps	Insert effect: MTAP	1–8	Number of echoes in the multi-tap pattern.
TB	Table	WAVE: WAVE (wavetable track)	list	The track's table (TABL).
TENS	Tension	Insert effect: SPRG	0–100	Spring tension: tighter and brighter, or looser and boingier.
TH	Threshold	WAVE: REC	level	Recording starts when the source passes this level.
THR	Threshold	Insert effect: CMP	−60…0 dB	Level above which the compressor works.
THR	Master compressor threshold	SETTINGS: MASTR	−60–0 dB	Level above which the master compressor works.
THRS	Threshold	Insert effect: DUCK	−60…0 dB	Level of the source above which the track ducks.
TILT	Tilt (harmonic model)	Macro: MACRO	0–100	First macro of the HARM model: spectral tilt.
TILT	Key tilt	Wavetable: KEYS	−600–600	Makes every key of the KEYS table darker (down) or brighter (up), by up to 6 dB per octave.
TIME	Strum time	ARP: PLAY	0–100 ms	Gap between strummed notes.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
TIME	Delay time	Insert effect: DLY, RVSE, TAPE, LCR, MTAP	varies	Time between echoes. With SYNC on it picks a tempo division.
TIME	Scan time	Wavetable: SCAN	0.05–60 s	Time one pass through the table takes, while SYNC is off.
TLEN	Time lane length	Wave seq: LANES	1–16	Length of the step-duration lane.
TN	Tune	DRUMS: bar	20 Hz–12 kHz	TUNE of the selected drum pad.
TO	Tone	DRUMS: bar	0–100	TONE of the selected drum pad.
TONE	Tone	Drums: DR	0–100	Brightness, in the way of each type: the click of FMKK, how far KICK's pitch falls, MKIK's resonance, the noise band of the snares, the width of CLAP's band, the high-pass of the hats and cymbal, TOM's stick noise, RIM's noise, a broad hiss to a pitched rattle on SHKR, CLAV's upper partial, sine to square on SUB, a low-pass on KSYN.
TONE	Tone	Insert effect: DRV, WSH, AMP	varies	Brightness of the effect.
TR	Transpose	SONG: bar	semitones	Transposes pitched tracks for this row.
TRIG	LFO trigger mode	FM, Macro, Wavetable, Wave seq, Sampler, Drums: LFO1 / LFO2	FREE KEY ONE HALF HOLD	FREE runs on; KEY restarts on each note; ONE plays one cycle then holds; HALF plays half a cycle; HOLD samples one value per note.
TRKB	Filter B key tracking	Synth: FLT B	0–100	How far filter B's cutoff follows the note.
TUN2	Oscillator 2 tune	Synth: OSC	−24…24 semitones	Pitch of oscillator 2 relative to oscillator 1, in semitones.
TUN3	Oscillator 3 tune	Synth: OSC3	−24…24 semitones	Pitch of oscillator 3 in semitones.
TUNE	Tune	Drums: DR	20 Hz–12 kHz	Pitch of the drum voice in Hz.
TUNE	Tune	Synth: OSC3; FM: RATIO; Macro: MACRO; Wavetable: OSC; Wave seq: SEQ	varies	Coarse pitch of the whole sound in semitones.
TUNE	Feedback tune	TRK FX: DIRT	30 Hz–4 kHz	Frequency the DIRT block's feedback path resonates at.
TY	Type	DRUMS: bar	14 models	TYPE of the selected drum pad.
TYPB	Filter B type	Synth: FLT B	SVF SEM LADR DIOD COMB OFF	Model of the second filter.
TYPE	Drum type	Drums: DR	15 values (App. C)	Synthesis model of this drum voice (15 models; Appendix C). KSYN, the kick machine, adds the KICK page.
TYPE	Filter type	Synth, FM, Macro, Wavetable, Wave seq, Sampler: FILTR; Insert effect: FLT	varies	Filter model: SVF, SEM, LADR (ladder), DIOD (diode), COMB, or OFF.
TYPE	Distortion type	TRK FX: DIRT	SOFT HARD FOLD ASYM	SOFT, HARD, FOLD (wavefolder) or ASYM (asymmetric).
TYPE	Insert effect type	TRK FX: FX	30 types	Effect in the selected slot. Changing it replaces the effect and its four values.
VAR	Variation	ARP: VARY	number	Selects a stored variation.
VDLY	Vibrato delay (CC 78)	MIDI track: SOUND	0–127	Sends the vibrato-delay controller.
VDPT	Vibrato depth (CC 77)	MIDI track: SOUND	0–127	Sends the vibrato-depth controller.
VEL	Strum velocity	ARP: PLAY	0–100	Velocity change across the strum.

SHORT	FULL NAME	BELONGS TO	VALUES	WHAT IT DOES
VEL	Step velocity	STEP: TRIG	0–127	Velocity of the step's notes.
VEL	Velocity to cutoff	Synth, FM, Macro, Wavetable, Wave seq: F ENV	0–100	Harder notes open the filter further.
VEL	Velocity sensitivity	Synth, Wavetable, Wave seq, Sampler: AMP	0–100	0: every note at full level. 100: level follows how hard you play.
VEL	Arp velocity lock	TRK FX: ARPLK	0–100	Per-step lock of the arpeggiator's velocity.
VELF	Velocity to filter	Sampler: FILTR	0–100	Harder notes open the filter further.
VELP	Velocity to position	Wavetable: WAVE	0–100	Harder notes start further into the table.
VIBR	Vibrato	Wavetable: OSC	0–100	Depth of pitch vibrato.
VL	Volume	PRESET: bar	0–100	Track volume, to level-match while browsing.
VOIC	Voices	Insert effect: CHR	1–3	Number of chorus voices.
VOL	Volume (CC 7)	MIDI track: MIDI	0–127	Sends the volume controller.
VOL	Track volume	TRACKS: VOL	0–100	Mixer page: knob N is the volume of track N.
VOWL	Vowel (formant model)	Macro: MACRO	0–100	First macro of the FORM model: vowel.
VOWL	Vowel	TRK FX: TALK	0–100	Position through the vowels A, E, I, O, U.
VOX	Voices	Synth: VOICE; Sampler: AMP	varies	Maximum number of notes that sound at once.
VRAT	Vibrato rate (CC 76)	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP; MIDI track: SOUND	varies	Sends the vibrato-rate controller.
WAV1	Oscillator 1 kind	Synth: OSC	10 values (App. C)	What oscillator 1 plays: an analog wave (SIN SAW SQR TRI PLS NOIS) or a wavetable (WT), FM pair (FM), formant (FORM) or harmonic (HARM) oscillator.
WAV2	Oscillator 2 kind	Synth: OSC	10 values (App. C)	What oscillator 2 plays; the same choices as oscillator 1.
WAV3	Oscillator 3 kind	Synth: OSC3	10 values (App. C)	What oscillator 3 (the sub oscillator) plays; the same choices as oscillator 1.
WDTH	Width	Insert effect: ENS	0–100	Stereo width of the ensemble.
WET	Insert wet level	TRK FX: FX	0–100	How much of the effect is heard; 0 takes it out.
WHL	Mod strip to cutoff	FM, Macro, Wavetable, Wave seq: F ENV	−100…+100	The mod strip opens (or, negative, closes) the filter.
WRAP	Wrap	Wavetable: KEYS	OFF ON	On: after the last key the KEYS table returns to the first, so a looped scan has no jump.
WSHP	Waveshape variation	Synth: ANALG	0–100	Random change of the waveform on every note: a component tolerance at low settings, a difference heard from note to note towards the top.
WVIB	Strip vibrato	FM, Macro, Wavetable, Wave seq, Sampler, Drums: STRIP	0–1 semitones	STRIP page: vibrato the mod strip brings in, in semitones at the top of the strip. 0 = none.
XFAD	Crossfade	Sampler: STRCH	0–100	Overlap between successive pieces of the stretch; in SLCE, how quickly a slice is faded when the next arrives.
XFAD	Crossfade	Wave seq: SEQ	0–100	How much each step's wave blends into the next.
XMOD	Cross-modulation	Synth: OSC	0–100	How strongly oscillator 2 frequency-modulates oscillator 1; clangorous, bell-like tones.
ZM	Zoom	WAVE: WAVE	factor	Zooms the waveform view.

Appendix B Gestures at a glance

TO...	DO THIS
Go to a screen	`Shift` + pad
Next / previous knob page	Knob-page buttons
Jump to a page	`Func` + top-row pad
Select a track	Hold ▲ or ▼ + top pad N; or `Shift` + ▲ / ▼; or top pad on `TRACKS`
Mute / solo a track	Hold ▲ or ▼ + bottom pad N / `Func` + bottom pad on `TRACKS`
Move a control on all tracks	Hold ▲ or ▼ + turn
See what Shift or Func does here	Hold `Shift` or `Func` alone; knob-page buttons turn the cards
Fine / coarse	`Shift` + turn / `Func` + turn
Add or remove a step	Tap a pad on `STEP`
Set a step's note	Hold the step + key
Lock a knob to a step	Hold the step + turn
Remove that lock	Hold the step + `>` + turn
Change pattern	Hold `>` + pad
Keep / reload the pattern	`>` + `Rec` / `>` + `Play`
Copy / clear / paste	`Func` + `Rec` / `Play` / `>` (a whole pattern: hold half a second)
Undo / redo	`Func` + `Shift` + `Play` / `Rec`
Record	Hold `Rec` + `Play`; again = free / quantized; `Play` stops and shows the steps
Fill	Hold `Shift` + `Rec`
Stop, rewind and silence everything	`Shift` + `Play`
Tap tempo	`Func` + `Settings`, tapped
Arp on / off	`Arp` (opens the screen), `Arp` again
Latch	`Shift` + `Arp`
Scale lock / scale screen	`Scale` / `Shift` + `Scale`
Pressure range on a knob	Turn it, then `Func` + ▲ (high end) or ▼ (low end); again clears
Pin the playing pads	On `DRUMS` or `SCALE` : `Shift` + hold that screen's bottom pad half a second
Confirm something destructive	Four clicks clockwise on its knob
Split the keyboard	`Shift` + `Oct−` / `Oct+`

Appendix C Value lists

LIST	VALUES, IN ORDER
Engines (`ENG`)	Synth · FM · Macro oscillator · Wavetable · Wave sequencer · Sampler · Drums · MIDI track
Oscillator waves	SIN SAW SQR TRI PLS NOIS WT FM FORM HARM
Filter types	SVF SEM LADR DIOD COMB OFF
LFO shapes (`SHPE`)	SIN TRI SQR SAW RND SRND RAMP EXPU EXPD
LFO trigger modes (`TRIG`)	FREE KEY ONE HALF HOLD
LFO divisions (`DIV`)	8BAR 4BAR 2BAR 1BAR 1/2D 1/2 1/4D 1/2T 1/4 1/8D 1/4T 1/8 1/16D 1/8T 1/16 1/16T 1/32 (D = dotted, T = triplet)
Effect sync divisions	1/128 1/64 1/32T 1/32 1/16T 1/32D 1/16 1/8T 1/16D 1/8 1/4T 1/8D 1/4 1/2T 1/4D 1/2 1/2D 1BAR 2BAR 4BAR 8BAR 16BAR
Insert effect types	CHR DRV FLG PHS TRM DLY RVB EQ CMP LMT FLT ENS PAN ROT WSH RED HALL SPRG SHIM RVSE TRNS RING VIB WAH AMP GEQ TAPE LCR MTAP DUCK
Drum types	FMKK KICK MKIK SNAR FMSN CLAP HAT OHAT CYMB TOM RIM SHKR CLAV SUB KSYN
Sampler play modes	FWD REV 1SHT GATE LOOP PING GRAN
Slice play modes	ONE KEYS NEXT RAND
Stretch algorithms	OFF TAPE CYCL GRAN SLCE SMER WSOL
Stretch lengths (bars)	1/16 1/8 1/4 1/2 1 2 4 8 16 32
Wavetable scan modes	OFF RUN LOOP PING STEP RAND
Wavetable frame options	64 128 256 A64 A128 A256
Grain shapes	HANN TRI PERC SWEL
Wavetables	SAW SQR TRI SIN PWM FORM HARM SPEC DIGI USER KEYS VOWL SYNC FOLD FM1 FM3 BELL ORGN NOIS RESO WLSH PRIM OCTV PDST COMB
Macro models	VA FM FOLD FORM HARM PLCK DUST BOW
FM algorithms	6>1 2+3 3x2 2x3 1>3 ADD 2+4 4+2
Trig conditions	ALL PROB A:B FILL !FIL 1ST !1ST PRE !PRE
Arp modes	Up; Down; Up-down; Random; As played; Chord repeat; Walk; Play order; Converge; Diverge; Poly arp (all notes, accented); Random with octave jumps
Arp rates	1/1 1/2 1/4 1/4T 1/8 1/8T 1/16 1/16T 1/32 1/32T 1/64 1/64T (the knob runs slowest to fastest)
Roll divisions	1/2 1/4 1/4T 1/8 1/8T 1/16 1/16T 1/32 1/32T 1/64 1/64T
Track speeds	1/4 1/2 1x 2x 4x
Scale pages (built in)	Chrom Major Minor Penta Blues Whole Dorian Mixo HrMin mPent Phryg Lydia
Preset categories	ALL INIT BASS PAD LEAD KEYS FX PERC KIT
Clock sources	INT EXT AUTO LINK

Appendix D Glossary of terms

TERM	MEANING
Bar screen	A screen with a picture or list and a strip of eight two-letter knob labels along the bottom.
Bus	One of two shared effects that every track can send to.
Cell	One of the eight positions on the display; the knob in the same position edits it.
Condition	A rule on a step that decides whether it plays on a given pass.
Engine	The sound generator of a track.
Fill	A held state in which steps marked FILL play.
Four clicks	The confirmation for anything destructive: four clockwise clicks on its knob.
Grain	A short slice of a sample, played many at a time in granular mode.
Grid screen	A screen with eight cells in two rows of four.
Insert	An effect in a slot on one track.
Knob page	One set of eight cells; the knob-page buttons move between them, going round at the ends.
Latch	Notes or an arp carrying on after the keys are released.
Lock (parameter lock)	A knob value that applies to one step only.
Micro-timing	A step's offset from the grid, up to half a step either way.
Name bar	The top line of the display.
Page key	Func + a top-row pad: jumps straight to a page.
Pattern	One of 16 sets of steps for all tracks.
Pin	Keeping the DRUMS or SCALE pads playable on every screen.
Pressure	Polyphonic aftertouch: how hard each pad is pressed after it is struck, per pad.
Pressure range	A range on a knob that a bottom-row pad sweeps with pressure.
Ratchet	Several quick repeats inside one step.
Resampling	Recording Neptune's own output into a new sample.
Roll	A pad repeating while held, faster with pressure.
Route	A pressure destination and depth on a track.
Row	One line of the song: a pattern, a repeat count and a transposition.
Send	The amount of a track fed to a bus.
Slice	One piece of a sample, cut evenly or at a hit.
Time stretch	Changing how long a sound lasts without changing its pitch.
Transient	The sudden start of a sound, such as a drum hit; where transient slicing cuts.
Step	One position in a track's pattern.

TERM	MEANING
Track layer	What is available while ▲ or ▼ is held: track select, mute, and all-track turns.
Trig	A step that is switched on.